

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\H2 Storage aftercooler-AK.pvdb

Date of Analysis : Sep 5, 2026 3:41pm

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 1220.000 mm. ID
 Vessel Design Length, Tangent to Tangent 7283.95 mm.
 Specified Datum Line Distance 50.80 mm.
 Shell Side Design Temperature 85 °C
 Channel Side Design Temperature 85 °C
 Shell Side Design Pressure 20.000 bars
 Channel Side Design Pressure 15.400 bars
 Shop Shell Side Test Pressure 34.586 bars
 Field Shell Side Test Pressure 40.492 bars
 Shop Channel Side Test Pressure 26.991 bars
 Field Channel Side Test Pressure 31.851 bars
 Wind Design Code ASCE/SEI 7-22
 Earthquake Design Code ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Flange	SA-266 2	...	...	K03506	
Nozzle	SA-266 2	...	...	K03506	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-179	...	...	K01200	
Tubesheet	SA-266 2	...	...	K03506	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
LHS Dish End	15.520	0.00	21.800	3.0000	No	No
LHS Ch Shell	15.520	0.00	24.400	3.0000	N/A	No
LHS Ch Flg	15.520	0.00	20.800	3.0000	N/A	No
main Shell-1	20.138	0.00	39.700	2.0000	N/A	No
main Shell-2	20.138	0.00	26.600	2.0000	N/A	No
main Shell-3	20.138	0.00	39.700	2.0000	N/A	No
RHS Ch Flg	15.520	0.00	26.600	2.0000	N/A	No
RHS Ch Shell	15.520	0.00	26.600	2.0000	N/A	No
RHS Dish End	15.520	1.03	21.800	3.0000	No	No

Liquid Level: 1220.00 mm. Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Ellipse	0.0	50.8	16.0	12.7	9.9	...	1.00	1.00



Cylinder	600.0	600.0	14.0	14.0	9.9	...	1.00	1.00
Body Flg	751.2	148.0	128.0	120.0	115.2	114.4	1.00	1.00
Cylinder	2841.2	2000.0	20.0	20.0	11.0	...	1.00	1.00
Cylinder	4741.2	1900.0	14.0	14.0	11.0	...	1.00	1.00
Cylinder	6726.2	1985.0	20.0	20.0	11.0	...	1.00	1.00
Body Flg	6982.3	163.0	143.0	135.0	130.0	130.0	1.00	1.00
Cylinder	7182.3	200.0	14.0	14.0	8.9	...	1.00	1.00
Ellipse	7233.1	50.8	16.0	12.7	9.9	6.2	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
T1-Tube Inlet	WRC-297	0.976	Shell	Passed
T2-Tube outlet	WRC-297	0.976	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	200.000	mm.
Saddle Bearing Angle	120.000	deg.
Centerline Dimension	1050.000	mm.
Wear Pad Width	380.000	mm.
Wear Pad Thickness	16.000	mm.
Wear Pad Bearing Angle	132.000	deg.
Distance from Saddle to Tangent	1450.000	mm.
Baseplate Length	1200.000	mm.
Baseplate Thickness	16.000	mm.
Baseplate Width	225.000	mm.
Number of Ribs (including outside ribs)	3	
Rib Thickness	16.000	mm.
Web Thickness	16.000	mm.
Height of Center Web	350.000	mm.
Number of Bolts in Baseplate	2	

Baseplate Sketch

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	16720.39	Kgf
Maximum Transverse Saddle Shear Load	1163.81	Kgf
Maximum Longitudinal Saddle Shear Load	2327.62	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	13398.76	Kgf
Maximum Transverse Saddle Shear Load	165.85	Kgf
Maximum Longitudinal Saddle Shear Load	105.49	Kgf

Weights:

Fabricated - Bare w/o removable internals	16016.6	kg.
Shop Test - Fabricated + water (full)	24197.5	kg.
Shipping - Fab. + rem. intls.+ shipping app.	16016.6	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	16016.6	kg.
Empty - Fab. + intls. + details + wghts.	16016.6	kg.
Operating - Empty + operating liquid (no ca)	24906.2	kg.
Field Test - Empty weight + test medium (full)	24197.5	kg.



Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
T1-Tube Inlet	24.4	...	...	OK	9.94	OK	Passed
T2-Tube outlet	24.4	...	...	OK	9.89	OK	Passed
S1-Shell inlet	34.4	...	...	OK	10.33	OK	Passed
S2-Shell outlet	34.3	...	...	OK	10.33	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 34.3 Nozzle : S2-Shell outlet [Shellside]
 Minimum MAWP Nozzles : 24.4 Nozzle : T1-Tube Inlet [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm.	Layout Angle deg	Dia. Limit mm.
20	T1-Tube Inlet	350.800	270.000	412.000
20	T2-Tube outlet	350.800	90.000	412.000
40	S1-Shell inlet	1591.975	90.000	1008.000
60	S2-Shell outlet	6091.975	270.000	1008.000

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
T1-Tube Inlet	200 mm	Actual	WNF	228.0	14.000	...	...	339	300
T2-Tube outlet	200 mm	Actual	WNF	228.0	14.000	...	...	339	300
S1-Shell inlet	500 mm	Actual	WNF	532.0	16.000	...	...	316	300
S2-Shell outlet	500 mm	Actual	WNF	532.0	16.000	...	...	316	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
T1-Tube Inl	SA-266 2	14.000	10.000	...	...	...
T2-Tube out	SA-266 2	12.700	10.000	...	...	...
S1-Shell in	SA-266 2	14.000	10.000	...	...	...
S2-Shell ou	SA-266 2	14.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
T1-Tube Inlet	300.000	270.0	300.00	0.00	LHS Ch Shell
T2-Tube outlet	300.000	90.0	300.00	0.00	LHS Ch Shell
S1-Shell inlet	1541.175	90.0	200.00	0.00	main Shell-1
S2-Shell outlet	6041.175	270.0	200.00	0.00	main Shell-3

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 16.0mm. THK X 1220.0mm. ID X 50.8mm.	SA-516 70
1	WELD NECK FLANGE: 128.0mm. THK X 1430.0mm. OD	SA-266 2
1	CYLINDER: 20.0mm. THK X 1220.0mm. ID X 2000.0mm.	SA-516 70
1	CYLINDER: 14.0mm. THK X 1220.0mm. ID X 1900.0mm.	SA-516 70
1	CYLINDER: 20.0mm. THK X 1220.0mm. ID X 1985.0mm.	SA-516 70
1	WELD NECK FLANGE: 143.0mm. THK X 1430.0mm. OD	SA-266 2
10	BAFFLES: 1192.0mm DIA X 8.0mm THK	SA-516 70
8	TIE RODS: 25mm DIA X 5500mm	SA-193 B7M
2	TUBESHEET: 1430.0mm X 90.0mm THK	SA-266 2
722	TUBES: 6069.0mm X 25.4mm DIA X 2.1mm THK	SA-179
2	SADDLE: 200.0mm X 120 DEG	SA-516 70
2	GASKET: 1290.0mm. OD X 1252.0mm. ID	...
120	BODY FLANGE BOLTS: 30.0mm. DIA	SA-193 B7
120	NUTS FOR BODY FLANGE BOLTS: 30.0mm. DIA	...
1	CYLINDRICAL SEGMENT 14.0mm. THK X 3048.0mm. X 600.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm. THK X 784.7mm. X 600.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm. THK X 3048.0mm. X 200.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm. THK X 784.7mm. X 200.0mm.	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
main Shell-1	[8]	B	-7	-26	-29	0.663	20.000	1.00	Yes
main Shell-2	[8]	B	-18	-18	-29	0.995	14.000	1.00	Yes
main Shell-3	[8]	B	-7	-26	-29	0.663	20.000	1.00	Yes
Left Saddle	[24]	B	-31	-48		0.663	16.000	1.00	Yes
New Saddle	[24]	B	-31	-48		0.663	16.000	1.00	Yes
S1-Shell inlet	[1]	B	-7	-40	-29	0.497	20.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-48					
S2-Shell outle	[1]	B	-7	-40	-29	0.501	20.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-48					
Tubesheet: SS	[13]	B	-21	-48	-29	0.434	22.500	1.00	Yes
Warmest MDMT:			-7	-18					
LHS Ch Flg	[11]	B	-18	-27	-29	0.844	14.000	1.00	No
RHS Ch Flg	[11]	B	-18	-36	-29	0.676	14.000	1.00	No
LHS Dish End	[10]	B	-21	-24	-29	0.946	12.700	1.00	No
LHS Dish End	[7]	B	-14	-30	-29	0.716	16.000	1.00	No
LHS Ch Shell	[8]	B	-18	-27	-29	0.846	14.000	1.00	No
RHS Ch Shell	[8]	B	-18	-31	-29	0.774	14.000	1.00	No
RHS Dish End	[10]	B	-21	-24	-29	0.946	12.700	1.00	No
RHS Dish End	[7]	B	-14	-30	-29	0.716	16.000	1.00	No
T1-Tube Inlet	[1]	B	-18	-39	-29	0.631	14.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
T2-Tube outlet	[1]	B	-18	-39	-29	0.626	14.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
Tubesheet: CS	[14]	B	-4	-47	-29	0.434	22.500	1.00	No
Bolting	[21]		-48						
Bolting	[21]		-48						
Warmest MDMT:			-4	-24					
Exchanger Side		Computed MDMT		Required MDMT		Pass/Fail			
Shell		-29.0		0.0		Pass			
Channel/Tube		-29.0		0.0		Pass			

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.



UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (bars):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	20	bars
Channel Side Design Pressure	15.4	bars
Shell Side Design Temperature	85.0	°C
Channel Side Design Temperature	85.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Lethal	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	0.0	°C
MDMT (CET), Tube Side	0.0	°C
User defined MAWP, Shell Side	0	bars
User defined MAWP, Channel Side	0	bars
User defined MAPnc, Shell Side	0	bars
User defined MAPnc, Channel Side	0	bars
User defined Test Pres., Shell Side	0	bars
User defined Test Pres., Channel Side	0	bars

Projection of Nozzle from Vessel Top	0	mm.
Projection of Nozzle from Vessel Bottom	0	mm.
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes



Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-22	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	[V] 180	Km/hr
Surface Roughness Category	D: Flat, unobstructed	
Type of Surface	Moderately Smooth	
Base Elevation	0	mm.
Percent Wind for Hydrotest	33.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment	H or Hh	0 mm.
Distance Upwind of Crest	Lh	0 mm.
Distance from Crest to the Vessel	x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm.

Seismic Design Code	ASCE/SEI 7-22	
Importance Factor	1.500	
Force Modification Factor R	3.000	
Site Class	D	
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Component resonance duct. fact.	[CAR]	0.000
Component strength factor	[Rpo]	0.000
Overstrength factor	[Omega0]	0.000
Response Acceleration Factor	[SMS]	0.326
Response Acceleration Factor	[SM1]	0.139
Building Importance Factor	[Ie]	1.000
Site Latitude		0.000 deg
Site Longitude		0.000 deg

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
--	-----



Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/9
Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	LHS Dish End
Distance "FROM" to "TO"	50.8 mm.
Inside Diameter	1220 mm.
Element Thickness	12.7 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	16 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	15.4 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm.
Yield Stress, Operating	242 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	1220 mm.
Liquid Density	0.0009984 kg./cm ³

Element#	2/9
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	LHS Ch Shell
Distance "FROM" to "TO"	600 mm.
Inside Diameter	1220 mm.
Element Thickness	14 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	14 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	15.4 bars
Design Temperature Internal Pressure	85 °C



Design External Pressure 0 bars
 Design Temperature External Pressure 85 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 20
 Detail Type Liquid
 Detail ID Liquid: 20
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Liquid 1220 mm.
 Liquid Density 0.0009984 kg./cm³

Element From Node 20
 Detail Type Nozzle
 Detail ID T1-Tube Inlet
 Dist. from "FROM" Node / Offset dist 300 mm.
 Nozzle Diameter 200 mm
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 270.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 65.161 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-266 2

Element From Node 20
 Detail Type Nozzle
 Detail ID T2-Tube outlet
 Dist. from "FROM" Node / Offset dist 300 mm.
 Nozzle Diameter 200 mm
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 65.161 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-266 2

Element# 3/9
 Element From Node 30
 Element To Node 40
 Element Type Flange
 Description LHS Ch Flg
 Distance "FROM" to "TO" 148 mm.
 Flange Inside Diameter 1220 mm.
 Element Thickness 120 mm.
 Internal Corrosion Allowance 3 mm.
 Nominal Thickness 128 mm.
 External Corrosion Allowance 0 mm.
 Design Internal Pressure 15.4 bars
 Design Temperature Internal Pressure 85 °C
 Design External Pressure 0 bars
 Design Temperature External Pressure 85 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-266 2
 Allowable Stress, Ambient 138 N./mm²
 Allowable Stress, Operating 138 N./mm²
 Allowable Stress, Hydrotest 225 N./mm²
 Material Density 0.00775 kg./cm³
 P Number Thickness 32.004 mm.
 Yield Stress, Operating 229.57 N./mm²
 UCS-66 Chart Curve Designation B
 External Pressure Chart Name CS-2
 UNS Number K03506
 Product Form Forgings



Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	1220 mm.
Liquid Density	0.0009984 kg./cm ³

Element#	4/9
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	main Shell-1
Distance "FROM" to "TO"	2000 mm.
Inside Diameter	1220 mm.
Element Thickness	20 mm.
Internal Corrosion Allowance	2 mm.
Nominal Thickness	20 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	20 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm.
Yield Stress, Operating	242 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	40
Detail Type	Saddle
Detail ID	Left Saddle
Dist. from "FROM" Node / Offset dist	1450 mm.
Width of Saddle	200 mm.
Height of Saddle at Bottom	1050 mm.
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm.
Width of Wear Plate	380 mm.
Thickness of Wear Plate	16 mm.
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.0
Moment Factor	3.0
Dimension E at base (optional)	0 mm.
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm.

Element From Node	40
Detail Type	Liquid
Detail ID	Liquid: 40
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	1220 mm.
Liquid Density	0.001148 kg./cm ³

Element From Node	40
Detail Type	Nozzle



Detail ID	S1-Shell inlet	
Dist. from "FROM" Node / Offset dist	700	mm.
Nozzle Diameter	500	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	321.09	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	

Element From Node	40	
Detail Type	Weight	
Detail ID	Fireproofing we	
Dist. from "FROM" Node / Offset dist	1000	mm.
Miscellaneous Weight	250	Kgf
Offset from Element Centerline	0	mm.

Element#	5/9	
Element From Node	50	
Element To Node	60	
Element Type	Cylinder	
Description	main Shell-2	
Distance "FROM" to "TO"	1900	mm.
Inside Diameter	1220	mm.
Element Thickness	14	mm.
Internal Corrosion Allowance	2	mm.
Nominal Thickness	14	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	20	bars
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	bars
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	1220	mm.
Liquid Density	0.001148	kg./cm ³

Element#	6/9	
Element From Node	60	
Element To Node	70	
Element Type	Cylinder	
Description	main Shell-3	
Distance "FROM" to "TO"	1985	mm.
Inside Diameter	1220	mm.
Element Thickness	20	mm.
Internal Corrosion Allowance	2	mm.
Nominal Thickness	20	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	20	bars
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	bars
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	



Element From Node	60	
Detail Type	Saddle	
Detail ID	New Saddle	
Dist. from "FROM" Node / Offset dist	530	mm.
Width of Saddle	200	mm.
Height of Saddle at Bottom	1050	mm.
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm.
Width of Wear Plate	380	mm.
Thickness of Wear Plate	16	mm.
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.1	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm.
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm.
Element From Node	60	
Detail Type	Liquid	
Detail ID	Liquid: 60	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	1220	mm.
Liquid Density	0.001148	kg./cm ³
Element From Node	60	
Detail Type	Nozzle	
Detail ID	S2-Shell outlet	
Dist. from "FROM" Node / Offset dist	1300	mm.
Nozzle Diameter	500	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	300.99	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	
Element From Node	60	
Detail Type	Weight	
Detail ID	fireproofing we	
Dist. from "FROM" Node / Offset dist	1000	mm.
Miscellaneous Weight	250	Kgf
Offset from Element Centerline	0	mm.

Element#	7/9	
Element From Node	70	
Element To Node	80	
Element Type	Flange	
Description	RHS Ch Flg	
Distance "FROM" to "TO"	163	mm.
Flange Inside Diameter	1220	mm.
Element Thickness	135	mm.
Internal Corrosion Allowance	2	mm.
Nominal Thickness	143	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	15.4	bars
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	bars
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	225	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm.



Yield Stress, Operating	229.57	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	70	
Detail Type	Liquid	
Detail ID	Liquid: 70	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	1220	mm.
Liquid Density	0.0009984	kg./cm ³

Element#	8/9	
Element From Node	80	
Element To Node	90	
Element Type	Cylinder	
Description	RHS Ch Shell	
Distance "FROM" to "TO"	200	mm.
Inside Diameter	1220	mm.
Element Thickness	14	mm.
Internal Corrosion Allowance	2	mm.
Nominal Thickness	14	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	15.4	bars
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	bars
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	234	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm.
Yield Stress, Operating	242	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	80	
Detail Type	Liquid	
Detail ID	Liquid: 60	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	1220	mm.
Liquid Density	0.0009984	kg./cm ³

Element#	9/9	
Element From Node	90	
Element To Node	100	
Element Type	Elliptical	
Description	RHS Dish End	
Distance "FROM" to "TO"	50.8	mm.
Inside Diameter	1220	mm.
Element Thickness	12.7	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	16	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	15.4	bars
Design Temperature Internal Pressure	85	°C



Design External Pressure	1.0342	bars
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	90	
Detail Type	Liquid	
Detail ID	Liquid: 70	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	1220	mm.
Liquid Density	0.0009984	kg./cm ³



XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
LHS Dish End		50.8	...	50.8	...
LHS Ch Shell		650.8	...	600	...
LHS Ch Flg		801.975	...	148	...
main Shell-1		2891.98	...	2000	...
main Shell-2		4791.98	...	1900	...
main Shell-3		6776.97	...	1985	...
RHS Ch Flg		7033.15	...	163	...
RHS Ch Shell		7233.15	...	200	...
RHS Dish End		7283.95	...	50.8	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flange Input Data Values

Description: LHS Ch Flg :

LHS Ch Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	15.52	bars
Design Temperature		85	°C
Internal Corrosion Allowance	ci	3.0000	mm.
External Corrosion Allowance	ce	0.0000	mm.
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1220.000	mm.
Flange Outside Diameter	A	1430.000	mm.
Flange Thickness	t	120.0000	mm.
Thickness of Hub at Small End	go	14.0000	mm.
Thickness of Hub at Large End	g1	20.0000	mm.
Length of Hub	h	28.0000	mm.
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	1360.000	mm.
Nominal Bolt Diameter	a	30.0000	mm.
Type of Threads		TEMA Metric	
Number of Bolts		60	
Flange Face Outside Diameter	Fod	1301.000	mm.
Flange Face Inside Diameter	Fid	1220.000	mm.
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1290.000	mm.
Gasket Inside Diameter	Gi	1252.000	mm.
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm.
Raised face depth	Fd	0.0000	mm.

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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (15.5 \cdot (1220/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 15.5) + Ca$$

$$= 9.9412 \text{ mm.}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 11) / (613 + 0.6 \cdot 11)$$

$$= 24.498 \text{ bars}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 14) / (610 + 0.6 \cdot 14)$$

$$= 31.240 \text{ bars}$$

Corroded Flange Thickness, tc	= T-ci	117.000 mm.
Corroded Flange ID,	Bcor = B+2*ci	1226.000 mm.
Corroded Large Hub,	g1Cor = g1-ci	17.000 mm.
Corroded Small Hub,	g0Cor = go-ci	11.000 mm.
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	50.000 mm.
Gasket Contact Width,	N = (Go - Gi)/2	19.000 mm.

Basic Gasket Width, $bo = N/2$ 9.500 mm.
 Effective Gasket Width, $b = Cb \cdot \sqrt{bo}$ 7.767 mm.
 Gasket Reaction Diameter, $G = Go - 2 \cdot b$ 1274.466 mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
 $= \pi/4 \cdot G^2 \cdot P$
 $= \pi/4 \cdot 1274^2 \cdot 15.5$
 $= 201893.891 \text{ Kgf}$

Contact Load on Gasket Surfaces [Hp or Wo,2]:
 $= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$
 $= 2 \cdot 7.77 \cdot \pi \cdot 1274 \cdot 3 \cdot 15.5$
 $= 29529.414 \text{ Kgf}$

Hydrostatic End Load at Flange ID [Hd]:
 $= \pi/4 \cdot Bcor^2 \cdot P$
 $= \pi/4 \cdot 1226^2 \cdot 15.5$
 $= 186830.391 \text{ Kgf}$

Pressure Force on Flange Face [Ht]:
 $= H - Hd$
 $= 201894 - 186830$
 $= 15063.503 \text{ Kgf}$

Design Bolt Load for the Operating Condition [Wo]:
 $= \max(Wo,1 + Wo,2 + H'p, 0)$
 $= \max(201894 + 29529 + 0, 0)$
 $= 231423.312 \text{ Kgf}$

Design Bolt Load for the Gasket Seating Condition [Wg]:
 $= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$
 $= 68.9 \cdot 7.77 \cdot \pi \cdot 1274 + 0 \cdot 0 \cdot 0$
 $= 218641.438 \text{ Kgf}$

Required Bolt Area [Am]:
 $= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$
 $= \max((231423 + 0 + 4 \cdot 0/1274)/172, 218641/172)$
 $= 131.952 \text{ cm}^2$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:
 $= 2a + 6t/(m + 0.5)$
 $= 2 \cdot 30 + 6 \cdot 117/(3 + 0.5)$
 $= 260.571 \text{ mm.}$

Actual Circumferential Bolt Spacing [Bs]:
 $= C \cdot \sin(\pi / n)$
 $= 1360 \cdot \sin(\pi/60)$
 $= 71.177 \text{ mm.}$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:
 $= \max(\sqrt{ Bs/(2a + t) }, 1)$
 $= \max(\sqrt{ 71.2/(2 \cdot 30 + 117) }, 1)$
 $= 1.0000$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	131.952	301.779	
Radial Distance between Hub and Bolts:	40.500	50.000	
Radial Distance between Bolts and Edge:	32.500	35.000	
Circ. Spacing between the Bolts:	69.900	71.177	260.571

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:
 $= Ab \cdot Sa/(y \cdot \pi(Go + Gi))$
 $= 302 \cdot 172/(68.9 \cdot \pi(1290 + 1252))$
 $= 9.426 \text{ mm.}$



Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa(Am + Ab) / 2$$

$$= 172 (132 + 302) / 2$$

$$= 380348.91 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 231423 - 201894$$

$$= 29529.42 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1360 - 1274) / 2$$

$$= 42.7669 \text{ mm.}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((1360 - 1226) / 2 + 42.8)$$

$$= 54.8834 \text{ mm.}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1360 - 1226 - 17) / 2$$

$$= 58.5000 \text{ mm.}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	186830.	58.5000	1.0000	10930.
Face Pressure, Mt	15064.	54.8834	1.0000	827.
Gasket Load, Mg	29529.	42.7669	1.0000	1263.
Gasket Seating, Matm	380349.	42.7669	1.0000	16267.

Total Moment for Operation, Mo 13019. Kg-m.
 Total Moment for Gasket seating, Mg 16267. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 116.129 mm.
 Hub Ratio, h/h0 = HL / H0 0.241
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.545

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.892
 Factor V per 4.16.5 0.386
 Factor f per 4.16.5 1.444

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 1430/1226$$

$$= 1.166$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K)) / (K^2 - 1))) / (K - 1)$$

$$= (0.66845 + (5.7169(1.17^2 \cdot \log_{10}(1.17)) / (1.17^2 - 1))) / (1.17 - 1)$$

$$= 12.685$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.17^2(1 + 8.55246 \cdot \log_{10}(1.17)) - 1) / ((1.0472 + 1.9448 \cdot 1.17^2) \cdot (1.17 - 1))$$

$$= 1.852$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$



$$= (1.17^2(1 + 8.55246 \cdot \log_{10}(1.17)) - 1) / (1.36136(1.17^2 - 1) \cdot (1.17 - 1))$$
$$= 13.940$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$
$$= (1.17^2 + 1) / (1.17^2 - 1)$$
$$= 6.548$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$
$$= 13.9 \cdot 11^2 \cdot 116 / 0.39$$
$$= 506911.312 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$
$$= 0.89 / 116$$
$$= 0.008 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3 / d$$
$$= (117 \cdot 0.0077 + 1) / 1.85 + 117^3 / 506911$$
$$= 4.185$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$
$$= (1.44 \cdot 13019 / 1226) / (4.18 \cdot 17^2)$$
$$= 124.32 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$
$$= (1.44 \cdot 16267 / 1226) / (4.18 \cdot 17^2)$$
$$= 155.33 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$
$$= (2.2 \cdot 13019 / 1226) / (4.18 \cdot 117^2)$$
$$= 4.00 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$
$$= (2.2 \cdot 16267 / 1226) / (4.18 \cdot 117^2)$$
$$= 4.99 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$
$$= (12.7 \cdot 13019 / (117^2 \cdot 1226)) - 6.55 \cdot 4$$
$$= 70.34 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$
$$= (12.7 \cdot 16267 / (117^2 \cdot 1226)) - 6.55 \cdot 4.99$$
$$= 87.88 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$
$$= (124 + \max(4, 70.3)) / 2$$
$$= 97.33 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$
$$= (155 + \max(4.99, 87.9)) / 2$$
$$= 121.60 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$
$$= 231423 / 302$$
$$= 75.21 \text{ N./mm}^2$$



Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (218641/302)$$

$$= 71.05 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	124.32	207.00	155.33	207.00
Radial Flange	4.00	138.00	4.99	138.00
Tangential Flange	70.34	138.00	87.88	138.00
Maximum Average	97.33	138.00	121.60	138.00
Bolting	75.21	172.00	71.05	172.00

Minimum Required Flange Thickness [Rigidity] 115.189 mm.

Estimated M.A.W.P. (Operating) 20.9 bars
 Estimated Finished Weight of Flange at given Thk. 420.8 kg.
 Estimated Unfinished Weight of Forging at given Thk 501.4 kg.

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 16267 / 1 \cdot 9807 \cdot 0.39 / (3.79 \cdot 201334 \cdot 11^2 \cdot 116 \cdot 0.3)$
 $= 0.999 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 13019 / 1 \cdot 9807 \cdot 0.39 / (3.79 \cdot 197675 \cdot 11^2 \cdot 116 \cdot 0.3)$
 $= 0.814 \text{ (should be } \leq 1)$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 16267 / 1 \cdot 9807 \cdot 0.39 / (4.18 \cdot 201334 \cdot 11^2 \cdot 116 \cdot 0.3)$
 $= 0.904 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 13019 / 1 \cdot 9807 \cdot 0.39 / (4.18 \cdot 197675 \cdot 11^2 \cdot 116 \cdot 0.3)$
 $= 0.737 \text{ (should be } \leq 1)$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.84, Temperature Reduction per Fig. UCS 66.1 = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -27 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	115.02	207.00	140.51	207.00
Radial Flange	5.08	138.00	6.21	138.00
Tangential Flange	59.15	138.00	72.26	138.00
Maximum Average	87.08	138.00	106.38	138.00
Bolting	75.21	172.00	71.05	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 103.099 mm.

PV Elite 28 Licensee: SL User
FileName : H2 Storage aftercooler-AK
Flg Calc [Int P]: LHS Ch Flg

Flng: 5 3:41pm

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Flange Input Data Values

Description: RHS Ch Flg :

RHS Ch Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	15.52	bars
Design Temperature		85	°C
Internal Corrosion Allowance	ci	2.0000	mm.
External Corrosion Allowance	ce	0.0000	mm.
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1220.000	mm.
Flange Outside Diameter	A	1430.000	mm.
Flange Thickness	t	135.0000	mm.
Thickness of Hub at Small End	go	14.0000	mm.
Thickness of Hub at Large End	g1	20.0000	mm.
Length of Hub	h	28.0000	mm.
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	1360.000	mm.
Nominal Bolt Diameter	a	30.0000	mm.
Type of Threads		TEMA Metric	
Number of Bolts		60	
Flange Face Outside Diameter	Fod	1301.000	mm.
Flange Face Inside Diameter	Fid	1220.000	mm.
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1290.000	mm.
Gasket Inside Diameter	Gi	1252.000	mm.
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm.
Raised face depth	Fd	0.0000	mm.

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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (15.5 \cdot (1220/2 + 2)) / (138 \cdot 1 - 0.6 \cdot 15.5) + Ca$$

$$= 8.9299 \text{ mm.}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 12) / (612 + 0.6 \cdot 12)$$

$$= 26.742 \text{ bars}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 14) / (610 + 0.6 \cdot 14)$$

$$= 31.240 \text{ bars}$$

Corroded Flange Thickness, tc	= T-ci	133.000 mm.
Corroded Flange ID,	Bcor = B+2*ci	1224.000 mm.
Corroded Large Hub,	g1Cor = g1-ci	18.000 mm.
Corroded Small Hub,	g0Cor = go-ci	12.000 mm.
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	50.000 mm.
Gasket Contact Width,	N = (Go - Gi)/2	19.000 mm.

Basic Gasket Width, $bo = N/2$ 9.500 mm.
 Effective Gasket Width, $b = Cb \cdot \sqrt{bo}$ 7.767 mm.
 Gasket Reaction Diameter, $G = Go - 2 \cdot b$ 1274.466 mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
 $= \pi/4 \cdot G^2 \cdot P$
 $= \pi/4 \cdot 1274^2 \cdot 15.5$
 $= 201893.891 \text{ Kgf}$

Contact Load on Gasket Surfaces [Hp or Wo,2]:
 $= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$
 $= 2 \cdot 7.77 \cdot \pi \cdot 1274 \cdot 3 \cdot 15.5$
 $= 29529.414 \text{ Kgf}$

Hydrostatic End Load at Flange ID [Hd]:
 $= \pi/4 \cdot Bcor^2 \cdot P$
 $= \pi/4 \cdot 1224^2 \cdot 15.5$
 $= 186221.344 \text{ Kgf}$

Pressure Force on Flange Face [Ht]:
 $= H - Hd$
 $= 201894 - 186221$
 $= 15672.546 \text{ Kgf}$

Design Bolt Load for the Operating Condition [Wo]:
 $= \max(Wo,1 + Wo,2 + H'p, 0)$
 $= \max(201894 + 29529 + 0, 0)$
 $= 231423.312 \text{ Kgf}$

Design Bolt Load for the Gasket Seating Condition [Wg]:
 $= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$
 $= 68.9 \cdot 7.77 \cdot \pi \cdot 1274 + 0 \cdot 0 \cdot 0$
 $= 218641.438 \text{ Kgf}$

Required Bolt Area [Am]:
 $= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$
 $= \max((231423 + 0 + 4 \cdot 0/1274)/172, 218641/172)$
 $= 131.952 \text{ cm}^2$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:
 $= 2a + 6t/(m + 0.5)$
 $= 2 \cdot 30 + 6 \cdot 133/(3 + 0.5)$
 $= 288.000 \text{ mm.}$

Actual Circumferential Bolt Spacing [Bs]:
 $= C \cdot \sin(\pi / n)$
 $= 1360 \cdot \sin(\pi/60)$
 $= 71.177 \text{ mm.}$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:
 $= \max(\sqrt{ Bs/(2a + t) }, 1)$
 $= \max(\sqrt{ 71.2/(2 \cdot 30 + 133) }, 1)$
 $= 1.0000$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	131.952	301.779	
Radial Distance between Hub and Bolts:	40.500	50.000	
Radial Distance between Bolts and Edge:	32.500	35.000	
Circ. Spacing between the Bolts:	69.900	71.177	288.000

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:
 $= Ab \cdot Sa/(y \cdot \pi(Go + Gi))$
 $= 302 \cdot 172/(68.9 \cdot \pi(1290 + 1252))$
 $= 9.426 \text{ mm.}$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172 \cdot 302$$

$$= 529274.50 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 231423 - 201894$$

$$= 29529.42 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1360 - 1274) / 2$$

$$= 42.7669 \text{ mm.}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + h_g)$$

$$= 0.5 ((1360 - 1224) / 2 + 42.8)$$

$$= 55.3834 \text{ mm.}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g_1) / 2$$

$$= (1360 - 1224 - 18) / 2$$

$$= 59.0000 \text{ mm.}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	186221.	59.0000	1.0000	10987.
Face Pressure, Mt	15673.	55.3834	1.0000	868.
Gasket Load, Mg	29529.	42.7669	1.0000	1263.
Gasket Seating, Matm	529274.	42.7669	1.0000	22636.

Total Moment for Operation, Mo 13118. Kg-m.
 Total Moment for Gasket seating, Mg 22636. Kg-m.

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_o \cdot C_{or}}$ 121.194 mm.
 Hub Ratio, $h/h_o = H_L / H_o$ 0.231
 Thickness Ratio, $g_1/g_o = (g_1 \cdot C_{or}) / (g_o \cdot C_{or})$ 1.500

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.895
 Factor V per 4.16.5 0.398
 Factor f per 4.16.5 1.395

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 1430 / 1224$$

$$= 1.168$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.17^2 \cdot \log_{10}(1.17))) / (1.17^2 - 1)) / (1.17 - 1)$$

$$= 12.555$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.17^2 (1 + 8.55246 \cdot \log_{10}(1.17)) - 1) / ((1.0472 + 1.9448 \cdot 1.17^2) \cdot (1.17 - 1))$$

$$= 1.852$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$



$$= (1.17^2(1 + 8.55246 \cdot \log_{10}(1.17)) - 1) / (1.36136(1.17^2 - 1) \cdot (1.17 - 1))$$
$$= 13.796$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$
$$= (1.17^2 + 1) / (1.17^2 - 1)$$
$$= 6.481$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$
$$= 13.8 \cdot 12^2 \cdot 121 / 0.4$$
$$= 604708.750 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$
$$= 0.89 / 121$$
$$= 0.007 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3 / d$$
$$= (133 \cdot 0.0074 + 1) / 1.85 + 133^3 / 604709$$
$$= 4.961$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot M_{op} / B_{cor}) / (L \cdot g_1^2)$$
$$= (1.4 \cdot 13118 / 1224) / (4.96 \cdot 18^2)$$
$$= 91.22 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot M_{atm} / B_{cor}) / (L \cdot g_1^2)$$
$$= (1.4 \cdot 22636 / 1224) / (4.96 \cdot 18^2)$$
$$= 157.40 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot M_{op} / B_{cor}) / (L \cdot t^2)$$
$$= (2.31 \cdot 13118 / 1224) / (4.96 \cdot 13^2)$$
$$= 2.77 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot M_{atm} / B_{cor}) / (L \cdot t^2)$$
$$= (2.31 \cdot 22636 / 1224) / (4.96 \cdot 13^2)$$
$$= 4.77 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot M_{op} / (t^2 \cdot B_{cor})) - Z \cdot S_{Ro}$$
$$= (12.6 \cdot 13118 / (13^2 \cdot 1224)) - 6.48 \cdot 2.77$$
$$= 56.67 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (Y \cdot M_{atm} / (t^2 \cdot B_{cor})) - Z \cdot S_{Ra}$$
$$= (12.6 \cdot 22636 / (13^2 \cdot 1224)) - 6.48 \cdot 4.77$$
$$= 97.79 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2$$
$$= (91.2 + \max(2.77, 56.7)) / 2$$
$$= 73.95 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$
$$= (157 + \max(4.77, 97.8)) / 2$$
$$= 127.60 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= W_{m1} / A_b$$
$$= 231423 / 302$$
$$= 75.21 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (218641/302)$$

$$= 71.05 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	91.22	207.00	157.40	207.00
Radial Flange	2.77	138.00	4.77	138.00
Tangential Flange	56.67	138.00	97.79	138.00
Maximum Average	73.95	138.00	127.60	138.00
Bolting	75.21	172.00	71.05	172.00

Minimum Required Flange Thickness 129.997 mm.

Estimated M.A.W.P. (Operating) 26.7 bars
 Estimated Finished Weight of Flange at given Thk. 471.7 kg.
 Estimated Unfinished Weight of Forging at given Thk 552.2 kg.

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 22636 / 1 \cdot 9807 \cdot 0.4 / (4.52 \cdot 201334 \cdot 12^2 \cdot 121 \cdot 0.3)$
 $= 0.967 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 13118 / 1 \cdot 9807 \cdot 0.4 / (4.52 \cdot 197675 \cdot 12^2 \cdot 121 \cdot 0.3)$
 $= 0.571 \text{ (should be } \leq 1)$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 22636 / 1 \cdot 9807 \cdot 0.4 / (4.96 \cdot 201334 \cdot 12^2 \cdot 121 \cdot 0.3)$
 $= 0.881 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 13118 / 1 \cdot 9807 \cdot 0.4 / (4.96 \cdot 197675 \cdot 12^2 \cdot 121 \cdot 0.3)$
 $= 0.520 \text{ (should be } \leq 1)$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.68, Temperature Reduction per Fig. UCS 66.1 = 18 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -36 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	88.08	207.00	149.73	207.00
Radial Flange	3.28	138.00	5.57	138.00
Tangential Flange	51.43	138.00	87.43	138.00
Maximum Average	69.76	138.00	118.58	138.00
Bolting	75.21	172.00	71.05	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 124.689 mm.

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FileName : H2 Storage aftercooler-AK
Flg Calc [Int P]: RHS Ch Flg

Flng: 6 3:41pm

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
LHS Dish End		15.52	16	3	1220	138	1
LHS Ch Shell		15.52	14	3	1220	138	2
LHS Ch Flg		15.52	128	3	1220	138	3
main Shell-1		20.138	20	2	1220	138	4
main Shell-2		20.138	14	2	1220	138	5
main Shell-3		20.138	20	2	1220	138	6
RHS Ch Flg		15.52	143	2	1220	138	7
RHS Ch Shell		15.52	14	2	1220	138	8
RHS Dish End		15.52	16	3	1220	138	9

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
LHS Dish End		15.4	21.8	28.7	12.7	9.85755	1
LHS Ch Shell		15.4	24.4	31.2	14	9.94134	2
LHS Ch Flg		15.4	20.8	24.6	120	115.189	3
main Shell-1		20	39.7	44.4	20	11.0102	4
main Shell-2		20	26.6	31.2	14	11.0102	5
main Shell-3		20	39.7	44.4	20	11.0102	6
RHS Ch Flg		15.4	26.6	30.7	135	129.997	7
RHS Ch Shell		15.4	26.6	31.2	14	8.92998	8
RHS Dish End		15.4	21.8	28.7	12.7	9.85755	9

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 26.6 bars
 Shell Side MAPnc = 31.2 bars
 Channel Side MAWP = 20.8 bars
 Channel Side MAPnc = 24.6 bars

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: LHS Dish End

Elliptical Head Design Information:					
Design Pressure	P	15.520 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		50.800 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1220.000 mm.	Actual Stress		97.6 N./mm ²
Spec. Min. Thk t _e , or s _a		12.7000 mm.	Spec. Nominal Thk.		16.000 mm.
Surface Area		18989.6 cm ²	SG of Contents		1.000
Element Volume		297131.8 cm ³	Weight of Contents		297.0 Kgf
Empty Weight		251.2 Kgf	Operating Weight		548.2 Kgf
			Comp. All. Ope. Stress		123.8 N./mm ²
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 15.250 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 7.625 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (15.5 \cdot 1226 \cdot 0.99) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 15.5)$$

$$= 6.8576 + 3.0000 = 9.8576 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.120 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 9.7) / (0.99 \cdot 1226 + 0.2 \cdot 9.7)$$

$$= 21.9 - 0.12 = 21.8 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 12.7) / (1 \cdot 1220 + 0.2 \cdot 12.7)$$

$$= 28.7 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (15.5 \cdot (0.99 \cdot 1226 + 0.2 \cdot 9.7)) / (2 \cdot 1 \cdot 9.7)$$

$$= 97.605 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (15.5 \cdot 613) / (138 \cdot 1 - 0.6 \cdot 15.5) + 3$$

$$= 9.941 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.120 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 13) / (613 + 0.6 \cdot 13)$$

$$= 28.9 - 0.12 = 28.8 \text{ bars}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (1226 / (2 \cdot 308))^2) / 6$$
$$= 0.993522$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.571 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 12.7, tr = 9.18, c = 3 mm., $E^* = 1$
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.95$, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-21 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-24 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 16, tr = 9.31, c = 3 mm., $E^* = 1$
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.72$, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-30 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	20.762 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	0 °C
Governing Thk.	16.000 mm.	Basic MDMT	-14 °C
Stress/Thk. Ratio	0.716	Temperature Difference	16 °C
UCS-66(c) red.	No	Computed Min. Temp.	-30 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 15.250 mm.
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 7.625 mm.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: LHS Ch Shell

Cylindrical Shell Design Information:					
Design Pressure	P	15.520 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		600.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1220.000 mm.	Actual Stress		87.4 N./mm ²
Spec. Min. Thk t _e , or s _a		14.0000 mm.	Spec. Nominal Thk.		14.000 mm.
Surface Area		23524.2 cm ²	SG of Contents		1.000
Element Volume		701517.8 cm ³	Weight of Contents		701.1 Kg _f
Empty Weight		252.4 Kg _f	Operating Weight		953.5 Kg _f
			Comp. All. Ope. Stress		127.1 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E \cdot 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (15.5 \cdot 613) / (138 \cdot 1 \cdot 0.6 \cdot 15.5)$$

$$= 6.9413 + 3.0000 = 9.9413 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.120 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 11) / (613 + 0.6 \cdot 11)$$

$$= 24.5 - 0.12 = 24.4 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 14) / (610 + 0.6 \cdot 14)$$

$$= 31.2 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (15.5 \cdot (613 + 0.6 \cdot 11)) / (1 \cdot 11)$$

$$= 87.424 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 1.135 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 14, tr = 9.31, c = 3 mm., E* = 1

Thickness Ratio = tr * E*/(tg - c) = 0.85, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -27 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	20.762 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	0 °C
Governing Thk.	14.000 mm.	Basic MDMT	-18 °C

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Stress/Thk. Ratio	0.846	Temperature Difference	9 °C
UCS-66(c) red.	No	Computed Min. Temp.	-27 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: main Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	20.138 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		2000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1220.000 mm.	Actual Stress		69.7 N./mm ²
Spec. Min. Thk t _e , or s _a		20.0000 mm.	Spec. Nominal Thk.		20.000 mm.
Surface Area		79168.1 cm ²	SG of Contents		1.150
Element Volume		1606575.8 cm ³	Weight of Contents		1846.4 Kg
Empty Weight		1207.7 Kg	Operating Weight		3054.1 Kg
			Comp. All. Ope. Stress		137.8 N./mm ²

Radiography Information:				
User Defined		User Defined		
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (20.1 \cdot 612) / (138 \cdot 1 - 0.6 \cdot 20.1)$$

$$= 9.0102 + 2.0000 = 11.0102 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.138 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 18) / (612 + 0.6 \cdot 18)$$

$$= 39.9 - 0.14 = 39.7 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 20) / (610 + 0.6 \cdot 20)$$

$$= 44.4 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (20.1 \cdot (612 + 0.6 \cdot 18)) / (1 \cdot 18)$$

$$= 69.681 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.613 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 20, t_r = 11.9, c = 2 mm., E* = 1

Thickness Ratio = t_r * E* / (t_g - c) = 0.66, Temp. Reduction = 19 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-26 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:	
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MDMT Curve	B	Press. at MDMT	26.604 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	0 °C
Governing Thk.	20.000 mm.	Basic MDMT	-7 °C
Stress/Thk. Ratio	0.663	Temperature Difference	19 °C
UCS-66(c) red.	No	Computed Min. Temp.	-26 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: main Shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	20.138 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		1900.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1220.000 mm.	Actual Stress		103.9 N./mm ²
Spec. Min. Thk t _e , or s _a		14.0000 mm.	Spec. Nominal Thk.		14.000 mm.
Surface Area		74493.5 cm ²	SG of Contents		1.150
Element Volume		1526247.0 cm ³	Weight of Contents		1754.1 Kg
Empty Weight		799.2 Kg	Operating Weight		2553.4 Kg
			Comp. All. Ope. Stress		129.2 N./mm ²

Radiography Information:				
User Defined		User Defined		
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (20.1 \cdot 612) / (138 \cdot 1 - 0.6 \cdot 20.1)$$

$$= 9.0102 + 2.0000 = 11.0102 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.138 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 12) / (612 + 0.6 \cdot 12)$$

$$= 26.7 - 0.14 = 26.6 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 14) / (610 + 0.6 \cdot 14)$$

$$= 31.2 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (20.1 \cdot (612 + 0.6 \cdot 12)) / (1 \cdot 12)$$

$$= 103.917 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.135 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 14, t_r = 11.9, c = 2 mm., E* = 1

Thickness Ratio = t_r * E* / (t_g - c) = 0.99, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-18 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:	
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MDMT Curve	B	Press. at MDMT	26.604 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	0 °C
Governing Thk.	14.000 mm.	Basic MDMT	-18 °C
Stress/Thk. Ratio	0.995	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-18 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: main Shell-3

Cylindrical Shell Design Information:					
Design Pressure	P	20.138 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		1985.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1220.000 mm.	Actual Stress		69.7 N./mm ²
Spec. Min. Thk t _e , or s _a		20.0000 mm.	Spec. Nominal Thk.		20.000 mm.
Surface Area		78574.4 cm ²	SG of Contents		1.150
Element Volume		1594526.5 cm ³	Weight of Contents		1832.6 Kg
Empty Weight		1198.6 Kg	Operating Weight		3031.2 Kg
			Comp. All. Ope. Stress		137.8 N./mm ²

Radiography Information:				
User Defined		User Defined		
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E \cdot 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (20.1 \cdot 612) / (138 \cdot 1 \cdot 0.6 \cdot 20.1)$$

$$= 9.0102 + 2.0000 = 11.0102 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.138 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 18) / (612 + 0.6 \cdot 18)$$

$$= 39.9 - 0.14 = 39.7 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 20) / (610 + 0.6 \cdot 20)$$

$$= 44.4 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (20.1 \cdot (612 + 0.6 \cdot 18)) / (1 \cdot 18)$$

$$= 69.681 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.613 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 20, t_r = 11.9, c = 2 mm., E* = 1

Thickness Ratio = t_r * E* / (t_g - c) = 0.66, Temp. Reduction = 19 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-26 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:	
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MDMT Curve	B	Press. at MDMT	26.604 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	0 °C
Governing Thk.	20.000 mm.	Basic MDMT	-7 °C
Stress/Thk. Ratio	0.663	Temperature Difference	19 °C
UCS-66(c) red.	No	Computed Min. Temp.	-26 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: RHS Ch Shell

Cylindrical Shell Design Information:					
Design Pressure	P	15.520 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		200.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1220.000 mm.	Actual Stress		80.1 N./mm ²
Spec. Min. Thk t _e , or s _a		14.0000 mm.	Spec. Nominal Thk.		14.000 mm.
Surface Area		7841.4 cm ²	SG of Contents		1.000
Element Volume		233839.2 cm ³	Weight of Contents		233.7 Kg
Empty Weight		84.1 Kg	Operating Weight		317.8 Kg
			Comp. All. Ope. Stress		129.2 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E \cdot 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (15.5 \cdot 612) / (138 \cdot 1 \cdot 0.6 \cdot 15.5)$$

$$= 6.9300 + 2.0000 = 8.9300 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.120 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 12) / (612 + 0.6 \cdot 12)$$

$$= 26.7 - 0.12 = 26.6 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 14) / (610 + 0.6 \cdot 14)$$

$$= 31.2 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (15.5 \cdot (612 + 0.6 \cdot 12)) / (1 \cdot 12)$$

$$= 80.087 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.135 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 14, t_r = 9.29, c = 2 mm., E* = 1

Thickness Ratio = t_r * E* / (t_g - c) = 0.77, Temp. Reduction = 13 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -31 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	20.762 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	0 °C
Governing Thk.	14.000 mm.	Basic MDMT	-18 °C

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Stress/Thk. Ratio	0.774	Temperature Difference	13 °C
UCS-66(c) red.	No	Computed Min. Temp.	-31 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: RHS Dish End

Elliptical Head Design Information:					
Design Pressure	P	15.520 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		50.800 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1220.000 mm.	Actual Stress		97.6 N./mm ²
Spec. Min. Thk t,e, or sa		12.7000 mm.	Spec. Nominal Thk.		16.000 mm.
Surface Area		18989.6 cm ²	SG of Contents		1.000
Element Volume		297131.8 cm ³	Weight of Contents		297.0 Kg _f
Empty Weight		251.2 Kg _f	Operating Weight		548.2 Kg _f
			Comp. All. Ope. Stress		129.2 N./mm ²
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 15.250 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 7.625 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (15.5 \cdot 1226 \cdot 0.99) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 15.5)$$

$$= 6.8576 + 3.0000 = 9.8576 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.120 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 9.7) / (0.99 \cdot 1226 + 0.2 \cdot 9.7)$$

$$= 21.9 - 0.12 = 21.8 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 12.7) / (1 \cdot 1220 + 0.2 \cdot 12.7)$$

$$= 28.7 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (15.5 \cdot (0.99 \cdot 1226 + 0.2 \cdot 9.7)) / (2 \cdot 1 \cdot 9.7)$$

$$= 97.605 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (15.5 \cdot 613) / (138 \cdot 1 - 0.6 \cdot 15.5) + 3$$

$$= 9.941 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.120 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 13) / (613 + 0.6 \cdot 13)$$

$$= 28.9 - 0.12 = 28.8 \text{ bars}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (1226 / (2 \cdot 308))^2) / 6$$

$$= 0.993522$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.571 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 12.7, tr = 9.18, c = 3 mm., $E^* = 1$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.95$, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-21 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-24 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 16, tr = 9.31, c = 3 mm., $E^* = 1$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.72$, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-30 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	20.762 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	0 °C
Governing Thk.	16.000 mm.	Basic MDMT	-14 °C
Stress/Thk. Ratio	0.716	Temperature Difference	16 °C
UCS-66(c) red.	No	Computed Min. Temp.	-30 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 15.250 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 7.625 mm.

EXTERNAL PRESSURE RESULTS for: RHS Dish End

Elliptical Head From 90 to 100 Ext. Chart: CS-2 at 85 °C

Elastic Modulus from Chart: CS-2 at 85 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
9.700	1245.40	128.39	0.0010818	87.68

$$MAEP = B / (K_0 \cdot Do / t) = 87.7 / (0.9 \cdot 128) = 7.59 \text{ bars}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
3.225	1245.40	386.22	0.0003596	35.95

$$MAEP = B / (K_0 \cdot Do / t) = 36 / (0.9 \cdot 386) = 1.03 \text{ bars}$$

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	34.586 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	26.000 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	40.492 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	29.265 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	28.600 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	28.600 bars
App 27-4	= M.A.W.P.	26.604 bars

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	26.991 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	20.020 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	31.851 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	22.838 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	22.022 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	22.022 bars
App 27-4	= M.A.W.P.	20.762 bars

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
LHS Dish End	130.5	234.0	0.558	27.11
LHS Ch Shell	119.8	234.0	0.512	27.11
main Shell-1	107.9	234.0	0.461	34.71
main Shell-2	153.3	234.0	0.655	34.71
main Shell-3	107.9	234.0	0.461	34.71
RHS Ch Shell	119.8	234.0	0.512	27.11
RHS Dish End	130.5	234.0	0.558	27.11

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
T1-Tube Inlet	Nozzle	138.00	138.00	1.000
T2-Tube outlet	Nozzle	138.00	138.00	1.000
S1-Shell inlet	Nozzle	138.00	138.00	1.000
S2-Shell outlet	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS Dish End	138.00	138.00	1.000
LHS Ch Shell	138.00	138.00	1.000
LHS Ch Flg	138.00	138.00	1.000
main Shell-1	138.00	138.00	1.000
main Shell-2	138.00	138.00	1.000
main Shell-3	138.00	138.00	1.000
RHS Ch Flg	138.00	138.00	1.000
RHS Ch Shell	138.00	138.00	1.000
RHS Dish End	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):



Description	Ambient	Operating	Ratio
Tube Material	92.40	92.40	1.000
Tubesheet Material	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
T1-Tube Inlet	20.99	225.00	0.093
T2-Tube outlet	20.99	225.00	0.093
S1-Shell inlet	56.31	225.00	0.250
S2-Shell outlet	56.31	225.00	0.250

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	34.586 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	26.000 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	40.492 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	29.265 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	28.600 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	28.600 bars
App 27-4	= M.A.W.P.	26.604 bars

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	26.991 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	20.020 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	31.851 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	22.838 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	22.022 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	22.022 bars
App 27-4	= M.A.W.P.	20.762 bars

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
LHS Dish End	201.1	234.0	0.859	31.97
LHS Ch Shell	180.1	234.0	0.770	31.97
main Shell-1	140.5	234.0	0.601	40.61
main Shell-2	209.6	234.0	0.896	40.61
main Shell-3	140.5	234.0	0.601	40.61
RHS Ch Shell	165.0	234.0	0.705	31.97
RHS Dish End	201.1	234.0	0.859	31.97

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
T1-Tube Inlet	Nozzle	138.00	138.00	1.000
T2-Tube outlet	Nozzle	138.00	138.00	1.000
S1-Shell inlet	Nozzle	138.00	138.00	1.000
S2-Shell outlet	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS Dish End	138.00	138.00	1.000
LHS Ch Shell	138.00	138.00	1.000
LHS Ch Flg	138.00	138.00	1.000
main Shell-1	138.00	138.00	1.000
main Shell-2	138.00	138.00	1.000
main Shell-3	138.00	138.00	1.000
RHS Ch Flg	138.00	138.00	1.000
RHS Ch Shell	138.00	138.00	1.000
RHS Dish End	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):



Description	Ambient	Operating	Ratio
Tube Material	92.40	92.40	1.000
Tubesheet Material	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
T1-Tube Inlet	24.76	225.00	0.110
T2-Tube outlet	24.76	225.00	0.110
S1-Shell inlet	65.90	225.00	0.293
S2-Shell outlet	65.90	225.00	0.293

Elements Suitable for Test Pressure.

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: main Shell-1

Shell Test Pressure		40.49 bars
Shell Thickness	ts	20.0000 mm.
Shell Internal Corrosion Allowance	cas	0.0000 mm.
Shell External Corrosion Allowance	casext	0.0000 mm.
Inside Diameter of Shell	Ds	1220.000 mm.
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Test Pressure		31.85 bars
Channel Thickness	tc	14.0000 mm.
Channel Internal Corrosion Allowance	cac	0.0000 mm.
Channel External Corrosion Allowance	cacext	0.0000 mm.
Inside Diameter of Channel	Dc	1220.000 mm.
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	722
Tube Wall Thickness	et	2.1100 mm.
Tube Outside Diameter	D	25.4000 mm.
Total Straight Tube Length	Lt	6069.00 mm.
Straight Tube Length (bet. inner tubsht faces) L		5889.00 mm.
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		92.40 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	180.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	38.1000 mm.
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm.
Groove Weld Leg	ag	2.1100 mm.
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	590.520 mm.
Largest Center-to-Center Tube Distance	Ul	0.0000 mm.
Length of Expanded Portion of Tube	ltx	0.0000 mm.
Tube-side pass partition groove depth	hg	6.0000 mm.
Area of Tube Layout	Ap	9184.1 cm ²
Perimeter of Tube Layout	Cp	3649.0 mm.

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf B

Tubesheet Design Metal Temperature T 21.11 °C
 Tubesheet Material SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03506
 Tubesheet Allowable Stress at Temperature S 138.00 N./mm²
 Tubesheet Allowable Stress at Ambient Tt 138.00 N./mm²
 Thickness of Tubesheet h 90.0000 mm.
 Tubesheet Corr. Allowance (Shell side) Cats 0.0000 mm.
 Tubesheet Corr. Allowance (Channel side) Catc 0.0000 mm.
 Tubesheet Outside Diameter A 1430.000 mm.

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped? No
 Area of the Untubed Lanes AL 0.0 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration l 823.000 mm.
 Tube End condition corresponding to Span (1) k 0.80
 Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes
 Mean Shell Metal Temp. along Shell len. Tsm 21.11 °C
 Mean Tube Metal Temp. along Tube length Ttm 21.11 °C
 Junction Stress Reduction option Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side Channel
 Flange Outside Diameter A 1430.000 mm.
 Flange Inside Diameter B 1220.000 mm.
 Flange Face Outside Diameter Fod 1301.000 mm.
 Flange Face Inside Diameter Fid 1220.000 mm.
 Gasket Outside Diameter Go 1290.000 mm.
 Gasket Inside Diameter Gi 1252.000 mm.
 Small end Hub thk. g0 14.0000 mm.
 Large end Hub thk. g1 20.0000 mm.
 Gasket Factor, m 3.00
 Gasket Design Seating Stress y 68.95 N./mm²
 Flange Facing Sketch Code Sketch 1a
 Gasket Thickness tg 3.1750 mm.
 Full face Gasket Flange Option Program Selects

Bolting Information:

Diameter of Bolt Circle C 1360.000 mm.
 Nominal Bolt Diameter dB 30.0000 mm.
 Thread Series TEMA Metric Thread
 Number of Bolts n 60
 Bolt Material SA-193 B7
 Bolt Allowable Stress At Temperature Sb 172.00 N./mm²
 Bolt Allowable Stress At Ambient Sa 172.00 N./mm²
 Weld between Flange and Shell/Channel 0.0000 mm.

Note: The full bolt load Sa*Ab, was used for all load cases.

Tubesheet Integral with Shell
 Tubesheet Extended as Flange Yes
 Thickness of Extended Portion of Tubesheet Tf 54.0000 mm.
 Is Bolt Load Transferred to the Tubesheet Yes

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (31.9 \cdot (25.4/2 - 0)) / (92.4 \cdot 1 + 0.4 \cdot 31.9)$
 $= 0.4319 + 0.0000 = 0.4319 \text{ mm.}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 21.11 °C
 Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	25.40	5889.00	12.04	50.0000	0.0075908	95.82

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 95.8) / (3 \cdot 12) = 106 \text{ bars}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.9290	25.40	5889.00	27.34	50.0000	0.0014713	83.05

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 83.1) / (3 \cdot 27.3) = 40.5 \text{ bars}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.9290 mm.
Allowable Internal Pressure at Corroded thickness	164.43 bars
Required Internal Design Pressure	31.85 bars
Allowable External Pressure at Corroded thickness	106.12 bars
Required External Design Pressure	40.49 bars
Required Thickness due to Shell Side pressure	0.9290 mm.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (G_{oc} - G_{ic}) / 2$	19.000 mm.
Basic Gasket Width,	$b_0 = N / 2$	9.500 mm.
Effective Gasket Width,	$b = \sqrt{b_0} * 2.5$	7.764 mm.
Gasket Reaction Diameter,	$G = G_o - 2.0 * b$	1274.472 mm.

Results for ASME Fixed Tubesheet Calculations for Configuration b:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [ρ]:
 $= l_{tx} / h$ (modified for corrosion if present)
 $= 0/90$
 $= 0.0000$ (must be $0 \leq \rho \leq 1$)

Effective tube hole diameter [d^*]:
 $= \max(dt - 2 * tt(EtT/E) * (StT/S) * (\rho); dt - 2 * tt)$
 $= \max(25.4 - 2 * 2.11 (202701232/202701232) * (92.4/138) * (0); 25.4 - 2 * 2.11)$
 $= 25.4000$ mm.

Equivalent outer tube limit circle diameter [Do]:
 $= 2 * r_o + dt$
 $= 2 * 591 + 25.4$
 $= 1206.440$ mm.

Determine the basic ligament efficiency for shear [μ]:
 $= (p - dt) / p$
 $= (38.1 - 25.4) / 38.1$
 $= 0.333$

Equivalent outer tube limit radius [ao]:
 $= Do / 2$
 $= 1206/2$
 $= 603.2200$ mm.

Effective tube pitch [p^*]:
 $= p / \sqrt{ (1 - 4 * \min(AL * CNVfactor; 4 * Do * p) / (\pi * Do^2)) }$
 $= 38.1 / \sqrt{ (1 - 4 * \min(0 * 100 ; 4 * 1206 * 38.1) / (\pi * 1206^2)) }$
 $= 38.1000$ mm.

Effective ligament efficiency for bending [μ^*]:
 $= (p^* - d^*) / p^*$
 $= (38.1 - 25.4) / 38.1$
 $= 0.3333$

Ratio [ρ_s]:
 $= a_s / a_o$
 $= 610/603$
 $= 1.0112$

Ratio [ρ_c]:
 $= a_c / a_o$
 $= 637/603$
 $= 1.0564$

Compute parameter [x_t]:
 $= 1 - Nt((dt - 2 * tt) / (2 * a_o))^2$
 $= 1 - 722 ((25.4 - 2 * 2.11) / (2 * 603))^2$
 $= 0.7775$

Determine parameter [x_s]:

$$= 1 - Nt(dt/(2 \cdot ao))^2$$

$$= 1 - 722 (25.4/(2 \cdot 603))^2$$

$$= 0.6800$$

Determine the value [h'g]:
 $= \max((hg - CATC); 0)$ (For pressure only cases)
 $= \max((6 - 0); 0)$
 $= 6.000 \text{ mm.}$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot ts(Ds + ts)Es/L$$

$$= \pi \cdot 20 (1220 + 20)202701232/5889$$

$$= 273472.0625 \text{ Kgf/mm.}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot tt(Dt - tt)Et/L$$

$$= \pi \cdot 2.11 (25.4 - 2.11)202701232/5889$$

$$= 541.8925 \text{ Kgf/mm.}$$

Stiffness Factor [Ks,t]:

$$= Ks / (Nt \cdot Kt)$$

$$= 273472/(722 \cdot 542)$$

$$= 0.69898$$

Rigidity Ratio [J]:

$$= 1 / (1 + Ks/Kt)$$

$$= 1/(1 + 273472/0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - vs^2))^{1/4}) / ((Ds + ts)ts)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1220 + 20)20)$$

$$= 0.0115 \text{ 1/mm.}$$

Shell coefficient [ks]:

$$= \beta_s \cdot Es \cdot ts^3 / (6(1 - vs^2))$$

$$= 0.012 \cdot 202701232 \cdot 20^3 / (6(1 - 0.3^2))$$

$$= 34283364.0000 \text{ bars} \cdot \text{mm.}^2$$

Shell coefficient [λ_s]:

$$= 6(Ds + ts) \cdot ks/h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2/2)$$

$$= 6(1220 + 20) \cdot 34283364/90^3 \cdot (1 + 90 \cdot 0.012 + (90 \cdot 0.012)^2/2)$$

$$= 902203.2500 \text{ bars}$$

Shell coefficient [δ_s]:

$$= Ds^2 / (4 \cdot Es \cdot ts) \cdot (1 - Ds \cdot vs / (2(Ds + ts)))$$

$$= 1220^2 / (4 \cdot 202701232 \cdot 20) \cdot (1 - 1220 \cdot 0.3 / (2(1220 + 20)))$$

$$= 0.78235\text{E-}01 \text{ mm./N./mm}^2$$

Intermediate parameters for Tubesheet Gasketed on the Channel Side:
 β_C , k_C , λ_C , δ_C are all = 0

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$h/p = 2.362205$; $\mu^* = 0.333333$; $E^* = 65896428 \text{ KPa.}$
 $E^*/E = 0.325091$; $\nu^* = 0.334167$

Note: As h/p (2.362) is > 2, data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [Xa]:

$$= ((24(1 - \nu^{*2})Nt \cdot Et \cdot tt(dt - tt)ao^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.33^2)722 \cdot 202701232 \cdot 2.11$$

$$(25.4 - 2.11)603^2) / (65896428 \cdot 5889 \cdot 90^3))^{1/4}$$

$$= 3.7475$$

Values from Table 4.18.3

$Z_d = 0.029215$; $Z_v = 0.072525$; $Z_m = 0.399799$
 $Z_a = 0.500376\text{E}+01$; $Z_w = 0.072525$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1430/1206$$

$$= 1.1853$$

Compute coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.33) / 65896428 \cdot (902203 + 0 + 202701232 \cdot \ln(1.19))$$

$$= 1.2598$$

Compute parameter [ϕ]:

$$= (1 + \nu^*) F$$

$$= (1 + 0.33) 1.26$$

$$= 1.6808$$

Compute parameter [Q1]:

$$= (\rho_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.01 - 1 - 1.68 \cdot 0.073) / (1 + 1.68 \cdot 0.4)$$

$$= -0.066185$$

Compute Parameter [Qz1]:

$$= (Z_d + Q1 \cdot Z_w) / 2 \cdot Xa^4$$

$$= (0.029 + -0.066 \cdot 0.073) / 2 \cdot 3.75^4$$

$$= 2.4076$$

Compute parameter [Qz2]:

$$= (Z_v + Q1 \cdot Z_m) / 2 \cdot Xa^4$$

$$= (0.073 + -0.066 \cdot 0.4) / 2 \cdot 3.75^4$$

$$= 4.5425$$

Compute parameter [U]:

$$= (Z_w + (\rho_s - 1) Z_m) Xa^4 / (1 + \phi \cdot Z_m)$$

$$= (0.073 + (1.01 - 1) 0.4) 3.75^4 / (1 + 1.68 \cdot 0.4)$$

$$= 9.0849$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [yb]:

$$= (G_c - C) / D_o \text{ (config b)}$$

$$= (1274 - 1360) / 1206 = -0.071$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm. (For Pressure only cases)}$$

Calculate parameter [ω_s]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$$

$$= 1.01 \cdot 34283364 \cdot 0.012 \cdot 0.07823506 (1 + 90 \cdot 0.012)$$

$$= 6383.9458 \text{ mm.}^2$$

Calculate parameter [ω^*s]:

$$= a o^2 (\rho_s^2 - 1) \cdot (\rho_s - 1) / 4 - \omega_s$$

$$= 603^2 (1.01^2 - 1) \cdot (1.01 - 1) / 4 - 6384$$

$$= -6360.8320 \text{ mm.}^2$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta C (C_m \Theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \Theta + h \cdot C_q \Theta \cdot h \cdot \beta C / 2)]$$

$$= 1.06 \cdot 0 [0.000 (0 + 0 \cdot 0 \cdot 90) - 0.000 (0/90 + 0 \cdot 0/2)]$$

$$= 0.0000 \text{ mm.}^2$$

Parameter [ω^*c]:

$$= a o^2 [(\rho_c^2 + 1) \cdot (\rho_c - 1) / 4 - (\rho_s - 1) / 2] - \omega_c$$

$$= 603^2 [(1.06^2 + 1) \cdot (1.06 - 1) / 4 - (1.01 - 1) / 2] - 0$$

$$= 8809.4990 \text{ mm.}^2$$

Pressure [P*S]:
 = 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [P*C]:
 = 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= Ps \{ xs + 2(1 - xs) vt + [2/Kst \cdot (Ds/Do)^2] vs - [(ps^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)] [(Dj^2 - (Ds)^2)/Do^2] \}$$

$$= 40.5 \{ 0.68 + 2(1 - 0.68)0.3 + [2/0.7 \cdot (1220/1206)^2]0.3 - [(1.01^2 - 1)/(1 \cdot 0.7)] - [(1 - 1)/(2 \cdot 1 \cdot 0.7)] [(0^2 - (1220)^2)/1206^2] \}$$

 = 69.5426 bars

Pressure [P't]:

$$= [xt + 2(1 - xt) vt + 1/(J \cdot Kst)] Pt$$

$$= [0.78 + 2(1 - 0.78)0.3 + 1/(1 \cdot 0.7)] 31.9$$

 = 74.5851 bars

Pressure [P_y]:

$$= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2)$$

$$= 722 \cdot 542 \cdot 0 / (3.14 \cdot 603^2)$$

 = 0.0000 bars

Pressure [P_w]:

$$= -\gamma b \cdot U \cdot W* / (2 \cdot \pi \cdot ao^2)$$

$$= -0.071 \cdot 9.08 \cdot 529274 / (2 \cdot \pi \cdot 603^2)$$

 = 14.6210 bars

Calculate the pressure [Prim]:

$$= -(U/ao^2) \cdot (\omega*s \cdot Ps - \omega*c \cdot Pt)$$

$$= -(9.08/603^2) \cdot (-6361 \cdot 40.5 - 8809 \cdot 31.9)$$

 = 13.4363 bars

Calculate the pressure [P_w]:

$$= (U/ao^2) \cdot (\omega*s \cdot P*s - \omega*c \cdot P*c)$$

$$= (9.08/603^2) \cdot (6384 \cdot 0 - 0 \cdot 0)$$

 = 0.0000 bars

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst(Qz1 + (ps - 1)Qz2)) \cdot (P's - P't + P_y + P_w + P_w + Prim)$$

$$= 0.1000E+01 \cdot 0.7 / (1 + 1 \cdot 0.7(2.41 + (1.01 - 1)4.54)) \cdot (69.5 - 74.6 + 0 + 0 + 14.6 + 13.4)$$

 = 5.9175 bars

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega*s \cdot Ps - \omega*c \cdot Pt) - (\omega*s \cdot P*s - \omega*c \cdot P*c)) + W* \cdot \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Zm)$$

$$= [((-6361 \cdot 40.5 - 8809 \cdot 31.9) - (6384 \cdot 0 - 0 \cdot 0)) \cdot 0.01 + 529274 \cdot -0.071 / (2 \cdot \pi)] / (1 + 1.68 \cdot 0.4)$$

 = -6853.9893 Kgf

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2)$$

$$= -0.066 + 2 \cdot -6854 / (5.92 \cdot 603^2)$$

 = -0.69048

Fm Value from Table 4.18.3 = 0.345238

Tube-sheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot Fm / \mu*) \cdot (2 \cdot ao / (H - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.35 / 0.33) \cdot (2 \cdot 603 / (90 - 6))^2 \cdot 5.92$$

 = 189.6476 N./mm²



Allowable Tubesheet Bending Stress [σ allowed]:
 $= 1.5 \cdot \beta T \cdot S_y = 1.5 \cdot 0.95 \cdot 250 = 356 \text{ N./mm}^2, 4.18.233$

Tubesheet Bending Stress - final thickness [σ]:
 $= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$
 $= (1.5 \cdot 0.23 / 0.33) \cdot (2 \cdot 603 / (63.3 - 6))^2 \cdot 7.76$
 $= 356.2410 \text{ N./mm}^2$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:
 $= h + C_{ats} + C_{atc}$
 $= 63.3 + 0 + 0$
 $= 63.3365 \text{ mm.}$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:
 $= (2 \cdot \mu \cdot h / a_o) \cdot (0.8 \cdot \beta T \cdot S_y)$
 $= (2 \cdot 0.33 \cdot 90 / 603) \cdot 190$
 $= 18.8982 \text{ N./mm}^2$

Shear Stress is not required to be computed; [P_e] $\leq$ τ limit

Tubesheet Shear Stress is probably low, use the following req. thk:

Tubesheet thickness (Incl. Corr.) = 13.5357 mm.
 Tubesheet Shear Stress = 35.5409 N./mm²

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :
 $= \max(H_{reqB}; H_{reqS})$
 $= \max(63.3; 13.5)$
 $= 63.3365 \text{ mm.}$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:
 $F_{tmin} = -2.06, F_{tmax} = 2.18$

First Extreme Tube Axial Stress from among all the tubes [σ_1]:
 $= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s)$
 $= ((40.5 \cdot 0.68 - 31.9 \cdot 0.78) - (5.92) \cdot -2.06) /$
 $(0.78 - 0.68)$
 $= 15.3258 \text{ N./mm}^2$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_2]:
 $= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s)$
 $= ((40.5 \cdot 0.68 - 31.9 \cdot 0.78) - (5.92) \cdot 2.18) /$
 $(0.78 - 0.68)$
 $= -10.3790 \text{ N./mm}^2$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:
 $= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 15.3 \text{ N./mm}^2$

Allowable Tube Stress [$\sigma_{t,A}$]:
 $= \beta T \cdot S_{y,t} = 170.9969 \text{ N./mm}^2$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:
 $= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25)$
 $= \max((3.25 - 0.25(0.029 + -0.69 \cdot 0.073)3.75^4); 1.25)$
 $= 2.0000 \text{ (Should be } \leq 2)$

Determine the factor [rt]:
 $= (\sqrt{dt^2 + (dt - 2 \cdot tt)^2}) / 4$
 $= (\sqrt{25.4^2 + (25.4 - 2 \cdot 2.11)^2}) / 4$
 $= 8.2680 \text{ mm.}$

Determine the factor [Ct]:
 $= \sqrt{(2 \cdot \pi^2 \cdot E_t / S_{y,t})}$
 $= \sqrt{(2 \cdot \pi^2 \cdot 202701232 / 180)}$



$$= 154.2491$$

Determine the factor [Ft]:

$$= k \cdot L/r$$

$$= 0.8 \cdot 823/8.27$$

$$= 79.6325$$

Buckling Allowable Stress [Stb]:

$$= (S_y/t/F_s) \cdot (1 - F_t/(2 \cdot C_t))$$

$$= (180/2) \cdot (1 - 79.6/(2 \cdot 154))$$

$$= 170.997 \text{ N./mm}^2 \text{ (Never greater than } S_{ot})$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$

$$= 15.3 \cdot 1.54$$

$$= 241.27 \text{ Kg}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 2.11) 2.11$$

$$= 1.544 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.11 (25.4 - 2.11) 171$$

$$= 2691.916 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 171$$

$$= 2634.3147 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 171/171$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 171$$

$$= 1704.5566 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2 \cdot \sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d) - 0.75 \cdot d_o}$$

$$= 2 \cdot \sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.11 \cdot (25.4 - 2.11) 1 \cdot 1) - 0.75 \cdot 25.4}$$

$$= 2.667 \text{ mm.}$$

$$\text{Minimum Required Fillet Weld Leg [afr]: } 1.3334 \text{ mm.}$$

$$\text{Minimum Required Groove Weld Leg [agr]: } 1.3334 \text{ mm.}$$

Tube-Tubesheet Jt allowable, 2692 is $\geq$ tube strength 2692 Kg

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= \beta T \cdot S_{y,s} = 0.95 \cdot 260 = 370 \text{ N./mm}^2$$

Axial Membrane Stress in Shell [σ_s ,m]:

$$= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 603^2 / ((1220 + 20) \cdot 20) \cdot [5.92 + (1.01^2 - 1) \cdot (40.5 - 31.9)] + 610^2 \cdot 31.9 / ((1220 + 20) \cdot 20)$$

$$= 56.7623 \text{ N./mm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 20 mm., the minimum Shell length adjacent to the tubesheet should be 281.169 mm.

Computations completed for ASME tubesheet configuration: b

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	189.6	$\leq 356.2 \text{ N./mm}^2$	Pass
Tubesheet Shear Stress	5.0	$\leq 190.0 \text{ N./mm}^2$	Pass
Maximum Tube Stress	15.3	$\leq 171.0 \text{ N./mm}^2$	Pass
Minimum Tube Stress (Buckling)	-10.4	$\leq -171.0 \text{ N./mm}^2$	Pass
Maximum Force on any one Tube	241.3	$\leq 2691.9 \text{ Kg}$	Pass
Axial Membrane Stress in Shell	56.8	$\leq 370.5 \text{ N./mm}^2$	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm.)	Required	Actual	P/F
Tubesheet Thickness :	63.337	90.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.333	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.333	2.110	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/ Fail
D1uc	47.818 37.328	151 356	11 190	Ps+Pt-Th D1	Ok
D2uc	41.278 ...	99 356	16 190	Ps+Pt-Th D2	Ok
D3uc	63.337 ...	190 356	5 190	Ps+Pt-Th D3	Ok
Max:	63.3365 37.328 mm.	0.532	0.086 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Shell Axial Membrane Stress Summary:



---- Shell Stresses ----					:	---- Shell Band Stress ----				:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	Fail
D1uc	27	370	...	...	:	...	...	...	...	:	Ok
D2uc	30	370	...	...	:	...	...	...	...	:	Ok
D3uc	57	370	...	...	:	...	...	...	...	:	Ok
Max RATIO		0.153		...		...		...			

Tube, Shell and Channel Stress Summary:

---- Tube Stresses Tube Loads Shell Stress Channel Stress Pass											
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	43	170	-43	-170	671	2691	...	370	...	...	Ok
D2uc	34	170	-14	-170	528	2691	...	370	...	...	Ok
D3uc	15	170	-10	-170	241	2691	...	370	...	...	Ok
Max RATIO	0.249		0.249		0.249		...		...		

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm.)	Required	Actual	P/F
Tubesheet thickness :	63.337	90.000	Ok
Tubesheet thickness flanged extension :	37.328	54.000	Ok
Tube thickness :	0.929	2.110	Ok
Tube-tubesheet fillet weld leg :	1.333	2.110	Ok
Tube-tubesheet groove weld leg :	1.333	2.110	Ok

Min Shell length of thk, (20.000) adj. to tubesheet: 281.169 mm.

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm.
 Due to Pressure Shell Compresses by : -0.174 mm.
 Due to Pressure + Thermal Shell Compresses by : -0.174 mm.

Summary of Exchanger Pressure and Thermal Displacements: (mm.)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0580	...	...	...	...	...
2	-0.1737	...	...	...	...	...
3	-0.1157	...	...	...	...	...
4	...	...	...	...	...	...
5	...	...	...	...	...	...
6	...	...	...	...	...	...
7	...	...	...	...	...	...
8	...	...	...	...	...	...

External Pressure Calculation Results Summary:

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External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	No Calc	1245.4	9.7	0.0010818	87.6794
20	30	900.467	1248	11	0.001587	98.2327
30	40	No Calc	...	117	No Calc	No Calc
40	50	5885	1260	18	0.0004408	44.07
50	60	5885	1248	12	0.00024163	24.1575
60	70	5885	1260	18	0.0004408	44.07
70	80	No Calc	...	133	No Calc	No Calc
80	90	515.467	1248	12	0.0033245	113.66
90	100	No Calc	1245.4	9.7	0.0010818	87.6794

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	12.7	No Calc	...	7.59
20	30	14	No Calc	...	11.5
30	40	120	114.351	...	No Calc
40	50	20	No Calc	...	8.39
50	60	14	No Calc	...	3.1
60	70	20	No Calc	...	8.39
70	80	135	129.997	...	No Calc
80	90	14	No Calc	...	14.6
90	100	12.7	6.22457	1.03419	7.59

Minimum

3.1

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	900.467	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	5885	No Calc	No Calc	No Calc
50	60	5885	No Calc	No Calc	No Calc
60	70	5885	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc
80	90	515.467	No Calc	No Calc	No Calc
90	100	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	251.201	297132	204.101	301242	25.1201
20	30	252.391	701518	198.79	708435	25.2391
30	40	447.943	183870	434.006	184193	44.7943
40	50	1207.71	1606575	1088.69	1621934	120.771
50	60	799.238	1526247	686.173	1540837	79.9238
60	70	1198.65	1594526	1080.52	1609770	119.865
70	80	498.756	201031	488.58	201246	49.8756
80	90	84.1304	233839	72.2286	235375	8.41304
90	100	251.201	297132	204.101	301242	25.1201
Total		4991	6641870.50	4457	6704275.50	499



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
10	Liqd	296.951	-101.667	...	...	Liquid: 10
20	Liqd	701.09	300	...	...	Liquid: 20
20	Nozl	71.6775	300	-710	...	T1-Tube Inlet
20	Nozl	71.6775	300	710	...	T2-Tube outlet
30	Liqd	183.758	74	...	...	Liquid: 30
40	Sadl	261.561	1450	812	...	Left Saddle
40	Liqd	1846.43	1000	...	...	Liquid: 40
40	Nozl	353.203	700	860	...	S1-Shell inlet
40	Wght	250	1000	...	...	Fireproofing we
50	Liqd	1754.11	950	...	...	Liquid: 40
60	Sadl	257.469	530	812	...	New Saddle
60	Liqd	1832.59	992.5	...	...	Liquid: 60
60	Nozl	331.091	1300	-860	...	S2-Shell outlet
60	Wght	250	1000	...	...	fireproofing we
70	Liqd	200.908	81.5	...	...	Liquid: 70
80	Liqd	233.697	100	...	...	Liquid: 60
90	Liqd	296.951	152.467	...	...	Liquid: 70
30	FTsh	951.619	196.175	...	...	Tubes Tubesheet Baffles Baffles Baffles Baffles Baffles Baffles Baffles Baffles Baffles Baffles Baffles Baffles Baffles
30	Tube	5767.38	3185.67	...	...	
30	RTsh	951.619	6175.17	...	...	
40	Bafl	100.794	824	...	...	
40	Bafl	100.794	1364	...	...	
40	Bafl	100.794	1904	...	...	
40	Bafl	100.794	2444	...	...	
40	Bafl	100.794	2984	...	...	
40	Bafl	100.794	3524	...	...	
40	Bafl	100.794	4064	...	...	
40	Bafl	100.794	4604	...	...	
40	Bafl	100.794	5144	...	...	
40	Bafl	101.792	5684	...	...	

Total Weight of Each Detail Type:

Saddles	519.0
Liquid	7346.5
Nozzles	827.6
Weights	500.0
Exchanger Components	8679.6
Liquid in Tubes	1543.1

Sum of the Detail Weights 19415.9 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	5490.3	5490.3	5490.3	5490.3	5490.3	5490.3
Saddles	519.0	519.0	519.0	519.0	519.0	519.0
Nozzles	827.6	827.6	827.6	827.6	827.6	827.6
Wld Weights	500.0	500.0	500.0	500.0	500.0	500.0
Exchanger	8679.6	8679.6	8679.6	8679.6	8679.6	8679.6
Ope. Liquid	...	...	...	...	...	7346.5
Tube Ope Lqd	...	...	...	...	...	1543.1
Test Liquid	...	6637.8	...	...	...	...
Tube Tst Lqd	...	1543.2	...	...	...	...
Totals	16016.6	24197.5	16016.6	16016.6	16016.6	24906.2

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	16016.6 kg.
Shop Test Wt.	- Fabricated weight + water (full)	24197.5 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	16016.6 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	16016.6 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	16016.6 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	24906.2 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	24318.8 kg.
Field Test Wt.	- Empty weight + test medium (full)	24197.5 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	1544094.5 cm ³
Weight of Ope Liq in tubes :	1543.1 kg.
Weight of Water in tubes :	1543.2 kg.

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	18989.6	18081.1
20	30	23524.2	22996.5
30	40	11229.5	5979.08
40	50	79168.1	76654.9
50	60	74493.5	72822.1
60	70	78574.4	76080
70	80	11903.3	6553.99
80	90	7841.42	7665.49
90	100	18989.6	18081.1

Total 324713.7 304914.1 cm²
 Total 349.5 328.2 ft²



Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
T1-Tube Inlet	47.63	51.00	85	300	GR 1.1	32.162	39.284	47.631	39.284
T2-Tube outlet	47.63	51.00	85	300	GR 1.1	31.737	39.709	47.631	39.709
S1-Shell inlet	47.63	51.00	85	300	GR 1.1	...	...	...	...
S2-Shell outle	47.63	51.00	85	300	GR 1.1	...	...	...	...

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 47.631 bars
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 51.000 bars

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 39.284 bars
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 51.000 bars

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm.	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kg	MAWP PD bars	Maximum Pressure Pmax bars	Pass Fail
T1-Tube Inlet	248.539	0.50	913	1224	20.762	39.284	Pass
T2-Tube outlet	248.539	0.50	900	1224	20.762	39.709	Pass

Evaluating Flange on Nozzle [T1-Tube Inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(913) + 4 \cdot \text{abs}(1224) \cdot 248.539 \leq \pi \cdot 248.539^3 \cdot [(47.631 - 20.762) + .500 \cdot 47.631]$$

$$= 0.15819E+05 \leq 0.24929E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle T1-Tube Inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(47.631, 0.500 \cdot 47.631 + 47.631 - 15819.0 / (\pi \cdot 248.539^3))$$

$$= \min(47.631, 39.284)$$

$$= 39.284 \text{ bars}$$

Evaluating Flange on Nozzle [T2-Tube outlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(900) + 4 \cdot \text{abs}(1224) \cdot 248.539 \leq \pi \cdot 248.539^3 \cdot [(47.631 - 20.762) + .500 \cdot 47.631]$$

$$= 0.15610E+05 \leq 0.24929E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle T2-Tube outlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(47.631, 0.500 \cdot 47.631 + 47.631 - 15609.9 / (\pi \cdot 248.539^3))$$

$$= \min(47.631, 39.709)$$

$$= 39.709 \text{ bars}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 Km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	Moderately Smooth
Base Elevation	0 mm.
Percent Wind for Hydrotect	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.19 \cdot 0.99) / (1 + 1.7 \cdot 3.4 \cdot 0.19)))$
 $= \min(0.85, 0.92)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.588
Structure Height to Diameter ratio	6.276

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
Exposure Category: D from Table 26.11-1
 $\alpha = 11.5$: $Zg = 589788$ mm.

Effective Height [z]:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1050 + 0 = 1050$ mm.
 $= 3.44$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z [Kz]:
Because z (3.44 ft.) < 15 ft.
 $= 2.41(15 / Zg)^{2/\alpha}$
 $= 2.41(15/1935)^{2/11.5}$
 $= 1.04$

Type of Hill: No Hill

Wind Directionality Factor [Kd]:
 $= 1$, per Table 26.6-1

As there is No Hill Present: [Kzt]:
 $K1 = 0$, $K2 = 0$, $K3 = 0$



Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0 \cdot 0 \cdot 0)^2$$

$$= 1$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.04 \cdot 1 \cdot 1 \cdot 1 \cdot 112^2)$$

$$= 33.1 \text{ psf } [162 \text{ Kgs/m}^2]$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 33.1 \cdot 0.85 \cdot 0.59 \cdot 4.16$$

$$= 68.9 \text{ lbs. } [31.3 \text{ Kgf}]$$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
LHS Dish End	1050.0	0.000	0.000	0.000	1.035	1.000	161.849
LHS Ch Shell	1050.0	0.000	0.000	0.000	1.035	1.000	161.849
LHS Ch Flg	1050.0	0.000	0.000	0.000	1.035	1.000	161.849
main Shell-1	1050.0	0.000	0.000	0.000	1.035	1.000	161.849
main Shell-2	1050.0	0.000	0.000	0.000	1.035	1.000	161.849
main Shell-3	1050.0	0.000	0.000	0.000	1.035	1.000	161.849
RHS Ch Flg	1050.0	0.000	0.000	0.000	1.035	1.000	161.849
RHS Ch Shell	1050.0	0.000	0.000	0.000	1.035	1.000	161.849
RHS Dish End	1050.0	0.000	0.000	0.000	1.035	1.000	161.849

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf
Fireproofing	0.00	1050.00	161.85	0.00
fireproofing	0.00	1050.00	161.85	0.00

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	1050	1494.48	3866.73	161.849	34.4029
20	30	1050	1497.6	8985.6	161.849	79.9462
30	40	1050	1464	2166.72	161.849	19.2776
40	50	1050	1512	30240	161.849	269.05
50	60	1050	1497.6	28454.4	161.849	253.163
60	70	1050	1512	30013.2	161.849	267.032
70	80	1050	1464	2386.32	161.849	21.2314
80	90	1050	1497.6	2995.2	161.849	26.6487
90	100	1050	1494.48	3866.73	161.849	34.4029

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-22	
Importance Factor		1.500	
Force Modification Factor R		3.000	
Site Class		D	
Component Elevation Ratio	z/h	0.000	
Amplification Factor	Ap	1.000	
Force Factor		0.000	
Consider Vertical Acceleration		Yes	
Minimum Acceleration Multiplier		0.000	
User Value of Sds (used if > 0)		0.000	
User Value of Sd1 (used if > 0)		0.000	
Component resonance duct. fact.	[CAR]	0.000	
Component strength factor	[Rpo]	0.000	
Overstrength factor	[Omega0]	0.000	
Response Acceleration Factor	[SMS]	0.326	
Response Acceleration Factor	[SM1]	0.139	
Building Importance Factor	[Ie]	1.000	
Site Latitude		0.000	deg
Site Longitude		0.000	deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.33 = 0.22$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.14 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(5.450.75)$
 = 0.071 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.071$, 0.03) per 12.8.2
 = 0.030
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.22 \cdot 23800 \cdot 1.5$
 = 2327.615 Kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.22 \cdot 23800$
 = 1034.5 Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	610	2163.61	211.601
20	30	610	2163.61	211.601
30	40	610	2163.61	211.601
40	50	610	2163.61	211.601
40	50	610	2163.61	211.601
50	60	610	2163.61	211.601



60	0	610	2163.61	211.601
60	70	610	2163.61	211.601
70	80	610	2163.61	211.601
80	90	610	2163.61	211.601
90	100	610	2163.61	211.601

The Earthquake Weight does not include the weight of the saddles.

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Shop/Field Installation Options :

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	3820.227 mm.
Center of Gravity of the Liquid	3598.936 mm.
Center of Gravity of the Nozzles	3177.165 mm.
Center of Gravity of the Added Weights (Operating)	3841.975 mm.
Center of Gravity of the Added Weights (Empty)	3841.975 mm.
Center of Gravity of the Tubesheet(s)	3836.475 mm.
Center of Gravity of the Tubes	3836.475 mm.
Center of Gravity of the Baffles	4148.379 mm.
Center of Gravity of Bare Shell New and Cold	3726.393 mm.
Center of Gravity of Bare Shell Corroded	3753.202 mm.
Vessel CG in the Operating Condition	3738.196 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	3783.963 mm.
Vessel CG in the Test Condition	3729.631 mm.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	621.00	mm.
Shell Thickness used in this Case	t	20.000	mm.
Stiffened Vessel Length per 4.15.6	L	5885.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	1450.00	mm.
Saddle Width	b1 or b	200.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	380.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1050.00	mm.
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		16458.83	Kgf
Pressure used in Analysis	P	20.069	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	51.33	138.00
Long. Stress at Bottom of Midspan	52.53	138.00
Long. Stress at Top of Saddles	57.15	138.00
Long. Stress at Bottom of Saddles	22.13	138.00
Tangential Shear in Shell	8.57	110.40
Circ. Stress at Horn of Saddle	14.32	172.50
Circ. Compressive Stress in Shell	0.99	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (1005/2 + 0) \cdot 1050/1119$
 $= 1414.9 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(320, 0) \cdot 1050/2980$
 $= 112.6 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(2328, 0) \cdot 1050/2980$
 $= 820.1 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (2328/2 + 0) \cdot 1050/1119$
 $= 3276.4 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 13182 + \max(113, 1415, 820, 3276)$$

$$= 16458.8 \text{ Kg}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587(0 + \pi/4(1.26 \cdot 1.2)^2)$$

$$= 3134.819 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	16720.39	Kgf
Transverse Shear Load	1163.81	Kgf
Longitudinal Shear Load	2327.62	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(1220/2 + 20 + 16)\sin(120/2), 2 \cdot 621)$$

$$= 1118.905 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -16459 \cdot 1450 [1 - (1 - 1450/5885 + (621^2 - 0^2)/(2 \cdot 1450 \cdot 5885)) / (1 + (4 \cdot 0)/(3 \cdot 5885))]$$

$$= -5341.0 \text{ Kg-m}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 16459 \cdot 5885/4((1 + 2(621^2 - 0^2)/(5885^2)) / (1 + (4 \cdot 0)/(3 \cdot 5885)) - 4 \cdot 1450/5885)$$

$$= 889.0 \text{ Kg-m}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 20.1 \cdot 621/(2 \cdot 12) - 889/(\pi \cdot 621^2 \cdot 12)$$

$$= 51.33 \text{ N/mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 20.1 \cdot 621/(2 \cdot 12) + 889/(\pi \cdot 621^2 \cdot 12)$$

$$= 52.53 \text{ N/mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 20.1 \cdot 621/(2 \cdot 18) - 5341/(0.11 \cdot \pi \cdot 621^2 \cdot 18)$$

$$= 57.15 \text{ N/mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm/(2t) + M1/(K1^* \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 20.1 \cdot 621/(2 \cdot 18) + 5341/(0.19 \cdot \pi \cdot 621^2 \cdot 18)$$

$$= 22.13 \text{ N/mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot hm/3))$$

$$= 16459(5885 - 2 \cdot 1450)/(5885 + (4 \cdot 0/3))$$

$$= 8348.3 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:



$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 8348 / (621 \cdot 18)$$

$$= 8.57 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(621 \cdot 18)}$$

$$= 82.466 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(200 + 1.56 \cdot \sqrt{(621 \cdot 18)}, 2 \cdot 1450)$$

$$= 364.93 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(Sr/S, 1)$$

$$= \min(138/138, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ6,r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 16459 \cdot 0.1 / (365(18 + 1 \cdot 16))$$

$$= -0.99 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, L ≥ 8Rm (4.15.28) [σ7,r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -16459 / (4(18 + 1 \cdot 16)365) -$$

$$3 \cdot 0.053 \cdot 16459 / (2(18 + 1 \cdot 16)^2)$$

$$= -14.32 \text{ N./mm}^2$$

Distance between Saddle Supports [Ls]:

$$= 2980.0 \text{ mm.}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot Ls(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.12005\text{E-}04 \cdot 2980(85 - 21.1)$$

$$= 2.286 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1200.0000	mm.
Baseplate Thickness	Bpthk	16.0000	mm.
Baseplate Width	Bpwid	225.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	16.0000	mm.
Web Thickness	Webtk	16.0000	mm.
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	242.0	N./mm ²
Height of Web at Center	Hw,c	350.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(612 \cdot 18)} + 380)18^3) / 12$$

$$= 26.425 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 26.4 + 97.9(125 - 9)^2$$

$$= 13197.743 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	543.7	18.0	9.0	97.9	88084.8	0.132E+05
Wearplate	380.0	16.0	26.0	60.8	158080.0	0.597E+04
Web	16.0	388.0	228.0	62.1	1415424.2	0.144E+05
BasePlate	225.0	16.0	430.0	36.0	1548000.1	0.335E+05



Totals	...	...	...	256.8	3209589.2	0.670E+05
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Distance to Centroid [C1]:

$$= AY / A$$

$$= 3209589/257$$

$$= 125.007 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 16459$$

$$= 3349.7297 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 2.0676 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 145.1984 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1050 - 612 \cdot \sin(120/2) / 1.05$$

$$= 543.880 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 1821.8872 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 3.3316 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 161.3315 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(16459 + 262) \cdot 225) / (4 \cdot 1200 \cdot 161)}$$

$$= 11.955 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (610 + 20 + 16)$$

$$= 1118.905 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1119 / (3 - 1)$$

$$= 559.452 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 559 \cdot 225/2$$

$$= 629.384 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 16720 / (1200 \cdot 225)$$

$$= 6.193 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 629 \cdot 6.19$$

$$= 3897.610 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 29.4 + 44.8$$



$$= 74.196 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 3898/74.2$$

$$= 5.152 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	200.0	...	32.0	...	0.107E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 200 - 16$$

$$= 184.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/74.2$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200/2$$

$$= 100.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1067/74.2)}$$

$$= 37.916 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1050 - \cos(120/2) (610 + 20 + 16) - 16$$

$$= 711.000 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008/242)}$$

$$= 127.710$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 711/37.9$$

$$= 18.752$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 2328 / (2 \cdot 1200) \cdot 559 \cdot 711/2$$

$$= 192.891 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($18.8 < 128$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (18.8)^2 / (2 \cdot 128^2)) 242 /$$

$$(5/3 + 3 \cdot (18.8) / (8 \cdot 128) - (18.8^3) / (8 \cdot 128^3))$$

$$= 139.1 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 5.15/139 + (193 \cdot 100/10666667) / 161$$

$$= 0.147$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
--	---	---	---	---	----	----

Rib	16.0	184.0	0.0	29.4	0.0	0.107E+04
Web	559.5	16.0	0.0	89.5	0.0	19.1
Totals	...	...	...	119.0	...	0.109E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/119$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200/2$$

$$= 100.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1050 - \cos(646/2)(16 + 0 + 0) - 0$$

$$= 388.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1085/119)}$$

$$= 30.204 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 388/30.2$$

$$= 12.846$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 2328/(2 \cdot 1200)$$

$$= 0.970 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.97 \cdot 559 \cdot 388$$

$$= 210.525 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 211/108522$$

$$= 19.024 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($12.8 < 128$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.8)^2 / (2 \cdot 128^2)) 242 / (5/3 + 3 \cdot (12.8) / (8 \cdot 128) - (12.8)^3 / (8 \cdot 128^3))$$

$$= 141.3 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 6.43/141 + (211 \cdot 100/1085) / 161$$

$$= 0.163$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	149.95 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	2.4000 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	1.6469 cm ²
Saddle Load Q0 (Weight)	Q0	13444.0 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	820.1 Kg



Maximum Transverse Force	Ft	1163.8 Kgf
Maximum Longitudinal Force	Fl	2327.6 Kgf
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:
 = 1190.9 Kgf

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
 = max(Ff, Fl) / (Bolt Area • Number of Bolts)
 = 2328 / (1.65 • 2)
 = 69.3 N./mm². Must be less than 99.9 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
 = Ft / (Bolt Area • Number of Bolts)
 = 1164 / (1.65 • 2)
 = 34.7 N./mm². Must be less than 99.9 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = Fl / (BoltShearAllowable • Nbolts)
 = 2328 / (99.9 • 2)
 = 1.1421 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = B • Ft + Sum of X Moments
 = 1050 • 1164 + 0
 = 1222.02 Kg-m.

Eccentricity (e):
 = Rmom / QO
 = 1222/13444
 = 90.90 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0, 1.14, 0]
 = 1.1421 cm²

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 16, c = 2 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.66, Temp. Reduction = 19 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	621.00	mm.
Shell Thickness used in this Case	t	20.000	mm.
Stiffened Vessel Length per 4.15.6	L	5885.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	1450.00	mm.
Saddle Width	b1 or b	200.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees



Wear Plate Width	b2 or b1	380.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²

Shell Allowable Stress used in Calculation	138.00	N./mm ²
Head Allowable Stress used in Calculation	138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle	1.00	
Circumferential Efficiency at Mid-Span	1.00	

Distance from Saddle Base to Centerline	B	1050.00	mm.
Coefficient of Friction	mu	0.10	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	14928.24	Kgf
Pressure used in Analysis	P	20.069 bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	51.39	138.00
Long. Stress at Bottom of Midspan	52.47	138.00
Long. Stress at Top of Saddles	55.05	138.00
Long. Stress at Bottom of Saddles	23.30	138.00
Tangential Shear in Shell	7.78	110.40
Circ. Stress at Horn of Saddle	12.99	172.50
Circ. Compressive Stress in Shell	0.90	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr}(F_t/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (1005/2 + 0) \cdot 1050/1119$
 $= 1414.9 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(320, 0) \cdot 1050/2980$
 $= 112.6 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(2328, 0) \cdot 1050/2980$
 $= 820.1 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr}(F_t/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (2328/2 + 0) \cdot 1050/1119$
 $= 3276.4 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 11652 + \max(113, 1415, 820, 3276)$
 $= 14928.2 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587 (0 + \pi/4 (1.26 \cdot 1.2)^2)$
 $= 3134.819 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	15185.71	Kgf
Transverse Shear Load	1163.81	Kgf
Longitudinal Shear Load	2327.62	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1



Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1220/2 + 20 + 16)\sin(120/2), 2 \cdot 621)$$

$$= 1118.905 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -14928 \cdot 1450 [1 - (1 - 1450/5885 + (621^2 - 0^2)/(2 \cdot 1450 \cdot 5885))/(1 + (4 \cdot 0)/(3 \cdot 5885))]$$

$$= -4844.3 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 [(1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L]$$

$$= 14928 \cdot 5885/4 [(1 + 2(621^2 - 0^2)/(5885^2))/(1 + (4 \cdot 0)/(3 \cdot 5885)) - 4 \cdot 1450/5885]$$

$$= 806.4 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 \cdot t)$$

$$= 20.1 \cdot 621/(2 \cdot 12) - 806/(\pi \cdot 621^2 \cdot 12)$$

$$= 51.39 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$$

$$= 20.1 \cdot 621/(2 \cdot 12) + 806/(\pi \cdot 621^2 \cdot 12)$$

$$= 52.47 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M_1/(K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 20.1 \cdot 621/(2 \cdot 18) - 4844/(0.11 \cdot \pi \cdot 621^2 \cdot 18)$$

$$= 55.05 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M_1/(K_1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 20.1 \cdot 621/(2 \cdot 18) + 4844/(0.19 \cdot \pi \cdot 621^2 \cdot 18)$$

$$= 23.30 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 14928(5885 - 2 \cdot 1450)/(5885 + (4 \cdot 0/3))$$

$$= 7571.9 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 7572 / (621 \cdot 18)$$

$$= 7.78 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(621 \cdot 18)}$$

$$= 82.466 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(200 + 1.56 \cdot \sqrt{(621 \cdot 18)}, 2 \cdot 1450)$$

$$= 364.93 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r/S, 1)$$

$$= \min(138/138, 1)$$

= 1.0000

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 14928 \cdot 0.1 / (365(18 + 1 \cdot 16))$$

$$= -0.90 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -14928 / (4(18 + 1 \cdot 16)365) -$$

$$3 \cdot 0.053 \cdot 14928 / (2(18 + 1 \cdot 16)^2)$$

$$= -12.99 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1200.0000	mm.
Baseplate Thickness	Bpthk	16.0000	mm.
Baseplate Width	Bpwid	200.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	16.0000	mm.
Web Thickness	Webtk	16.0000	mm.
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	242.0	N./mm ²
Height of Web at Center	Hw,c	350.0	mm.
Friction Coefficient	mu	0.100	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(612 \cdot 18)} + 380)18^3) / 12$$

$$= 26.425 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 26.4 + 97.9(120 - 9)^2$$

$$= 12124.499 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	543.7	18.0	9.0	97.9	88084.8	0.121E+05
Wearplate	380.0	16.0	26.0	60.8	158080.0	0.541E+04
Web	16.0	388.0	228.0	62.1	1415424.2	0.150E+05
BasePlate	200.0	16.0	430.0	32.0	1376000.1	0.307E+05
Totals	...	...	...	252.8	3037589.2	0.633E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 3037589 / 253$$

$$= 120.181 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 14928$$

$$= 3038.2197 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.9238 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 145.1984 \text{ N./mm}^2$$



Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$
 $= 1050 - 612 \cdot \sin(120/2) / 1.05$
 $= 543.880 \text{ mm.}$

Bending Moment, $M = F_h \cdot d = 1652.4598 \text{ Kg-m.}$

Bending Stress, $S_b = (M \cdot C_1 / I) = 3.0787 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 161.3315 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$
 $= \sqrt{3(14928 + 257) \cdot 200 / (4 \cdot 1200 \cdot 161)}$
 $= 10.742 \text{ mm.}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120/2) (610 + 20 + 16)$
 $= 1118.905 \text{ mm.}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 1119 / (3 - 1)$
 $= 559.452 \text{ mm.}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 559 \cdot 200 / 2$
 $= 559.452 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 15186 / (1200 \cdot 200)$
 $= 6.327 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 559 \cdot 6.33$
 $= 3539.866 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 29.4 + 44.8$
 $= 74.196 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 3540 / 74.2$
 $= 4.679 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	200.0	...	32.0	...	0.107E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 200 - 16$
 $= 184.000 \text{ mm.}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0 / 74.2$
 $= 0.000 \text{ mm.}$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200/2$$

$$= 100.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1067/74.2)}$$

$$= 37.916 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2)(\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1050 - \cos(120/2)(610 + 20 + 16) - 16$$

$$= 711.000 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008/242)}$$

$$= 127.710$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 711/37.9$$

$$= 18.752$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 2328 / (2 \cdot 1200) \cdot 559 \cdot 711/2$$

$$= 192.891 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($18.8 < 128$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (18.8)^2 / (2 \cdot 128^2)) 242 / (5/3 + 3 \cdot (18.8) / (8 \cdot 128) - (18.8)^3 / (8 \cdot 128^3))$$

$$= 139.1 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 4.68/139 + (193 \cdot 100/10666667) / 161$$

$$= 0.144$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	184.0	0.0	29.4	0.0	0.107E+04
Web	559.5	16.0	0.0	89.5	0.0	19.1
Totals	...	...	...	119.0	...	0.109E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/119$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200/2$$

$$= 100.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1050 - \cos(646/2)(16 + 0 + 0) - 0$$

$$= 388.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1085/119)}$$

$$= 30.204 \text{ mm.}$$



Slenderness ratio $[KL/r]$:

$$= KL/r$$

$$= 1 \cdot 388/30.2$$

$$= 12.846$$

Unit Force $[Force, u]$:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 2328 / (2 \cdot 1200)$$

$$= 0.970 \text{ Kg/mm.}$$

Moment at base of inner Rib $[M_{base, c}]$:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.97 \cdot 559 \cdot 388$$

$$= 210.525 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load $[\sigma_B, \text{base}, c]$:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 211/108522$$

$$= 19.024 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($12.8 < 128$) per AISC E2-1 $[S_{ca}]$:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.8)^2 / (2 \cdot 128^2)) 242 / (5/3 + 3 \cdot (12.8) / (8 \cdot 128) - (12.8)^3 / (8 \cdot 128^3))$$

$$= 141.3 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base, c} \cdot C1/I) / S_{ba}$$

$$= 6.04/141 + (211 \cdot 100/1085) / 161$$

$$= 0.161$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	149.95 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	2.4000 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	1.6469 cm ²
Saddle Load QO (Weight)	QO	11909.3 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	820.1 Kg
Maximum Transverse Force	Ft	1163.8 Kg
Maximum Longitudinal Force	Fl	2327.6 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction $[\tau_b, l]$:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 2328 / (1.65 \cdot 2)$$

$$= 69.3 \text{ N./mm}^2. \text{ Must be less than } 99.9 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction $[\tau_b, t]$:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1164 / (1.65 \cdot 2)$$

$$= 34.7 \text{ N./mm}^2. \text{ Must be less than } 99.9 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load $[Bltarearl]$:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load $[Bltarears]$:

$$= FI / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 2328 / (99.9 \cdot 2)$$

$$= 1.1421 \text{ cm}^2$$



Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1050 \cdot 1164 + 0$$

$$= 1222.02 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 1222/11909$$

$$= 102.61 \text{ mm.} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 1.14, 0]$$

$$= 1.1421 \text{ cm}^2$$

MDMT Calculations for: New Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, $t_g = 16$, $c = 2 \text{ mm.}$, $E^* = 1$

Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.66$, Temp. Reduction = $19 \text{ }^\circ\text{C}$

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B $-14 \text{ }^\circ\text{C}$

Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit $-31 \text{ }^\circ\text{C}$

Min Metal Temp. at Required thickness (UCS 66.1) $-48 \text{ }^\circ\text{C}$

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	620.00	mm.
Shell Thickness used in this Case	t	20.000	mm.
Stiffened Vessel Length per 4.15.6	L	5885.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	1450.00	mm.
Saddle Width	b1 or b	200.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	380.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1050.00	mm.
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		13137.19	Kgf
Pressure used in Analysis	P	40.553	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	89.39	234.00
Long. Stress at Bottom of Midspan	90.21	234.00
Long. Stress at Top of Saddles	79.10	234.00
Long. Stress at Bottom of Saddles	53.86	234.00
Tangential Shear in Shell	6.17	138.58
Circ. Stress at Horn of Saddle	10.27	350.99
Circ. Compressive Stress in Shell	0.73	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (332/2 + 0) \cdot 1050/1119$
 $= 466.9 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl, wind; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(105, 0) \cdot 1050/2980$
 $= 37.2 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$
 $= 12670 + \max(37.2, 467, 0, 0)$
 $= 13137.2 \text{ Kgf}$

Longitudinal Wind Force [FI]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587 (0 + \pi/4 (1.26 \cdot 1.2)^2)$
 $= 3134.819 \text{ N}$



Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	13398.76	Kgf
Transverse Shear Load	165.85	Kgf
Longitudinal Shear Load	105.49	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1220/2 + 20 + 16)\sin(120/2), 2 \cdot 620)$$

$$= 1118.905 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -13137 \cdot 1450 [1 - (1 - 1450/5885 + (620^2 - 0^2)/(2 \cdot 1450 \cdot 5885))/(1 + (4 \cdot 0)/(3 \cdot 5885))]$$

$$= -4264.5 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 \{ (1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L \}$$

$$= 13137 \cdot 5885/4 \{ (1 + 2(620^2 - 0^2)/(5885^2))/(1 + (4 \cdot 0)/(3 \cdot 5885)) - 4 \cdot 1450/5885 \}$$

$$= 708.2 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 40.6 \cdot 620/(2 \cdot 14) - 708/(\pi \cdot 620^2 \cdot 14)$$

$$= 89.39 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 40.6 \cdot 620/(2 \cdot 14) + 708/(\pi \cdot 620^2 \cdot 14)$$

$$= 90.21 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 40.6 \cdot 620/(2 \cdot 20) - 4264/(0.11 \cdot \pi \cdot 620^2 \cdot 20)$$

$$= 79.10 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 40.6 \cdot 620/(2 \cdot 20) + 4264/(0.19 \cdot \pi \cdot 620^2 \cdot 20)$$

$$= 53.86 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4h_m/3))$$

$$= 13137(5885 - 2 \cdot 1450)/(5885 + (4 \cdot 0/3))$$

$$= 6663.5 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 6663 / (620 \cdot 20)$$

$$= 6.17 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{620 \cdot 20}$$

$$= 86.857 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(200 + 1.56 \cdot \sqrt{(620 \cdot 20)}, 2 \cdot 1450)$$

$$= 373.71 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:
 = 1.0000 Materials are the same, test case

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 13137 \cdot 0.1 / (374 (20 + 1 \cdot 16))$$

$$= -0.73 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -13137 / (4(20 + 1 \cdot 16)374) -$$

$$3 \cdot 0.053 \cdot 13137 / (2(20 + 1 \cdot 16)^2)$$

$$= -10.27 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1200.0000	mm.
Baseplate Thickness	Bpthk	16.0000	mm.
Baseplate Width	Bpwid	225.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	16.0000	mm.
Web Thickness	Webtk	16.0000	mm.
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	350.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(610 \cdot 20)} + 380)20^3) / 12$$

$$= 36.820 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 36.8 + 110 (121 - 10)^2$$

$$= 13689.687 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	552.3	20.0	10.0	110.5	110461.5	0.137E+05
Wearplate	380.0	16.0	28.0	60.8	170240.0	0.529E+04
Web	16.0	388.0	230.0	62.1	1427840.0	0.151E+05
BasePlate	225.0	16.0	432.0	36.0	1555199.9	0.348E+05
Totals	...	...	...	269.3	3263741.5	0.689E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 3263741 / 269$$

$$= 121.175 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 13137$$

$$= 2673.7036 \text{ Kgf}$$

$$\begin{aligned} \text{Tension Stress, } St &= (Fh/As) = 1.6503 \text{ N./mm}^2 \\ \text{Allowed Stress, } Sa &= 0.6 * \text{Yield Str} = 155.9972 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} \text{Saddle Splitting Dimension [d]:} \\ &= B - R \cdot \sin(\theta/2)/(\theta/2 \text{ in radians}) \\ &= 1050 - 610 \cdot \sin(120/2)/1.05 \\ &= 545.534 \text{ mm.} \end{aligned}$$

$$\text{Bending Moment, } M = Fh * d = 1458.6254 \text{ Kg-m.}$$

$$\begin{aligned} \text{Bending Stress, } Sb &= (M * C1 / I) = 2.5153 \text{ N./mm}^2 \\ \text{Allowed Stress, } Sa &= 2/3 * \text{Yield Str} = 173.3302 \text{ N./mm}^2 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned} &= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 * \text{BasePlateLength} * \text{AllStress})} \\ &= \sqrt{3(13137 + 262)225 / (4 * 1200 * 173)} \\ &= 10.325 \text{ mm.} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

$$\begin{aligned} \text{Web Length Dimension [Web Length]:} \\ &= 2 * \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 * \cos(90 - 120/2) (610 + 20 + 16) \\ &= 1118.905 \text{ mm.} \end{aligned}$$

$$\begin{aligned} \text{Distance between Ribs [e]:} \\ &= \text{Web Length} / (\text{Nr ribs} - 1) \\ &= 1119 / (3 - 1) \\ &= 559.452 \text{ mm.} \end{aligned}$$

$$\begin{aligned} \text{Baseplate Pressure Area [Ap]:} \\ &= e * \text{Bpwid} / 2 \\ &= 559 * 225 / 2 \\ &= 629.384 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Bearing Pressure [Bp]:} \\ &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} * \text{BasePlateWidth}) \\ &= 13399 / (1200 * 225) \\ &= 4.963 \text{ Kg/cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Axial Load [P]:} \\ &= Ap * Bp \\ &= 629 * 4.96 \\ &= 3123.319 \text{ Kg} \end{aligned}$$

$$\begin{aligned} \text{Area of the Rib and Web [Ar]:} \\ &= \text{Rib Area} + \text{Web Area} \\ &= 29.4 + 44.8 \\ &= 74.196 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Compressive Stress [Sc]:} \\ &= P/Ar \\ &= 3123/74.2 \\ &= 4.128 \text{ N./mm}^2 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	200.0	...	32.0	...	0.107E+04

$$\begin{aligned} \text{Rib dimension [D]:} \\ &= \text{Saddle Width} - \text{Web Thickness} \\ &= 200 - 16 \\ &= 184.000 \text{ mm.} \end{aligned}$$

Distance to Centroid from Datum [ytot]:



$$= AY / A$$

$$= 0/74.2$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200/2$$

$$= 100.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1067/74.2)}$$

$$= 37.916 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1050 - \cos(120/2) (610 + 20 + 16) - 16$$

$$= 711.000 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008/260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 711/37.9$$

$$= 18.752$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 105 / (2 \cdot 1200) \cdot 559 \cdot 711/2$$

$$= 8.742 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($18.8 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (18.8)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (18.8) / (8 \cdot 123) - (18.8)^3 / (8 \cdot 123^3))$$

$$= 149.1 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 4.13/149 + (8.74 \cdot 100/10666667) / 173$$

$$= 0.032$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	184.0	0.0	29.4	0.0	0.107E+04
Web	559.5	16.0	0.0	89.5	0.0	19.1
Totals	...	...	...	119.0	...	0.109E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/119$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200/2$$

$$= 100.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1050 - \cos(646/2) (16 + 0 + 0) - 0$$

$$= 388.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{1085/119}$$

$$= 30.204 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 388/30.2$$

$$= 12.846$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 105/(2 \cdot 1200)$$

$$= 0.044 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.044 \cdot 559 \cdot 388$$

$$= 9.541 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 9.54/108522$$

$$= 0.862 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($12.8 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.8)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (12.8) / (8 \cdot 123) - (12.8)^3 / (8 \cdot 123^3))$$

$$= 151.6 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (M_{base,c} \cdot C_1/I) / S_{ba}$$

$$= 5.15/152 + (9.54 \cdot 100/1085) / 173$$

$$= 0.039$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	2.4000 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	1.6469 cm ²
Saddle Load Q0 (Weight)	Q0	12931.8 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	37.2 Kg
Maximum Transverse Force	Ft	165.9 Kg
Maximum Longitudinal Force	Fl	105.5 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 105/(1.65 \cdot 2)$$

$$= 3.1 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 166/(1.65 \cdot 2)$$

$$= 4.9 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q_0 > Q_L \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= FI / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 105/(103 \cdot 2)$$

= 0.0501 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1050 \cdot 166 + 0$$

$$= 174.15 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 174/12932$$

$$= 13.47 \text{ mm.} < Bplen/6 \rightarrow \text{No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 0.05, 0]$$

$$= 0.0501 \text{ cm}^2$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	620.00	mm.
Shell Thickness used in this Case	t	20.000	mm.
Stiffened Vessel Length per 4.15.6	L	5885.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	1450.00	mm.
Saddle Width	b1 or b	200.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	380.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1050.00	mm.
Coefficient of Friction	mu	0.10	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		11475.14	Kgf
Pressure used in Analysis	P	40.553	bars

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		89.44	234.00
Long. Stress at Bottom of Midspan		90.16	234.00
Long. Stress at Top of Saddles		77.05	234.00
Long. Stress at Bottom of Saddles		55.00	234.00
Tangential Shear in Shell		5.39	138.58
Circ. Stress at Horn of Saddle		8.98	350.99
Circ. Compressive Stress in Shell		0.64	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num of Saddles + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (332/2 + 0) \cdot 1050/1119$$

$$= 466.9 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(105, 0) \cdot 1050/2980$$

$$= 37.2 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 11008 + \max(37.2, 467, 0, 0)$$

$$= 11475.1 \text{ Kgf}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587 (0 + \pi/4 (1.26 \cdot 1.2)^2)$$

$$= 3134.819 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	11732.61	Kgf
Transverse Shear Load	165.85	Kgf
Longitudinal Shear Load	105.49	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1220/2 + 20 + 16) \sin(120/2), 2 \cdot 620)$$

$$= 1118.905 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -11475 \cdot 1450 [1 - (1 - 1450/5885 + (620^2 - 0^2) / (2 \cdot 1450 \cdot 5885)) / (1 + (4 \cdot 0) / (3 \cdot 5885))]$$

$$= -3725.0 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 11475 \cdot 5885 / 4 ((1 + 2(620^2 - 0^2)/(5885^2)) / (1 + (4 \cdot 0) / (3 \cdot 5885)) - 4 \cdot 1450/5885)$$

$$= 618.6 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$$

$$= 40.6 \cdot 620 / (2 \cdot 14) - 619 / (\pi \cdot 620^2 \cdot 14)$$

$$= 89.44 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 40.6 \cdot 620 / (2 \cdot 14) + 619 / (\pi \cdot 620^2 \cdot 14)$$

$$= 90.16 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 40.6 \cdot 620 / (2 \cdot 20) - 3725 / (0.11 \cdot \pi \cdot 620^2 \cdot 20)$$

$$= 77.05 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M_1 / (K_1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 40.6 \cdot 620 / (2 \cdot 20) + 3725 / (0.19 \cdot \pi \cdot 620^2 \cdot 20)$$



$$= 55.00 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a)/(L+(4 \cdot hm/3))$$

$$= 11475 (5885 - 2 \cdot 1450)/(5885 + (4 \cdot 0/3))$$

$$= 5820.4 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 5820/(620 \cdot 20)$$

$$= 5.39 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(620 \cdot 20)}$$

$$= 86.857 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(200 + 1.56 \cdot \sqrt{(620 \cdot 20)}, 2 \cdot 1450)$$

$$= 373.71 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1 (t + \eta \cdot tr))$$

$$= -0.76 \cdot 11475 \cdot 0.1/(374 (20 + 1 \cdot 16))$$

$$= -0.64 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q/(4(t+\eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q/(2(t+\eta \cdot tr)^2)$$

$$= -11475/(4(20 + 1 \cdot 16)374) -$$

$$3 \cdot 0.053 \cdot 11475/(2(20 + 1 \cdot 16)^2)$$

$$= -8.98 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1200.0000	mm.
Baseplate Thickness	Bpthk	16.0000	mm.
Baseplate Width	Bpwid	200.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	16.0000	mm.
Web Thickness	Webtk	16.0000	mm.
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	350.0	mm.
Friction Coefficient	mu	0.100	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3)/12$$

$$= ((1.56 \cdot \sqrt{(610 \cdot 20)} + 380) 20^3)/12$$

$$= 36.820 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C_1 - Y)^2$$

$$= 36.8 + 110 (116 - 10)^2$$

$$= 12563.083 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	552.3	20.0	10.0	110.5	110461.5	0.126E+05
Wearplate	380.0	16.0	28.0	60.8	170240.0	0.477E+04
Web	16.0	388.0	230.0	62.1	1427840.0	0.158E+05
BasePlate	200.0	16.0	432.0	32.0	1382399.9	0.319E+05
Totals	...	...	...	265.3	3090941.2	0.650E+05



Distance to Centroid [C1]:

$$\begin{aligned} &= AY / A \\ &= 3090941/265 \\ &= 116.489 \text{ mm.} \end{aligned}$$

Angle [β]:

$$\begin{aligned} &= 180 - \text{Saddle Angle}/2 \\ &= 180 - 120/2 \\ &= 120.0 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned} &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\ &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\ &= 0.2035 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned} &= K1 \cdot Q \\ &= 0.2 \cdot 11475 \\ &= 2335.4395 \text{ Kgf} \end{aligned}$$

$$\begin{aligned} \text{Tension Stress, } St &= (Fh / As) = 1.4788 \text{ N./mm}^2 \\ \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned} &= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians}) \\ &= 1050 - 610 \cdot \sin(120/2) / 1.05 \\ &= 545.534 \text{ mm.} \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 1274.0870 \text{ Kg-m.}$$

$$\begin{aligned} \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 2.2397 \text{ N./mm}^2 \\ \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned} &= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})) \\ &= \sqrt{(3(11475 + 257)200) / (4 \cdot 1200 \cdot 173)} \\ &= 9.109 \text{ mm.} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned} &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 \cdot \cos(90 - 120/2) (610 + 20 + 16) \\ &= 1118.905 \text{ mm.} \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned} &= \text{Web Length} / (\text{Nr ribs} - 1) \\ &= 1119 / (3 - 1) \\ &= 559.452 \text{ mm.} \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot \text{Bpwid} / 2 \\ &= 559 \cdot 200 / 2 \\ &= 559.452 \text{ cm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\ &= 11733 / (1200 \cdot 200) \\ &= 4.889 \text{ Kgf/cm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= Ap \cdot Bp \\ &= 559 \cdot 4.89 \\ &= 2734.931 \text{ Kgf} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= \text{Rib Area} + \text{Web Area} \\ &= 29.4 + 44.8 \\ &= 74.196 \text{ cm}^2 \end{aligned}$$



Compressive Stress [Sc]:

$$= P/A_r$$

$$= 2735/74.2$$

$$= 3.615 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	200.0	...	32.0	...	0.107E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 200 - 16$$

$$= 184.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/74.2$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200/2$$

$$= 100.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1067/74.2)}$$

$$= 37.916 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1050 - \cos(120/2) (610 + 20 + 16) - 16$$

$$= 711.000 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008/260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 711/37.9$$

$$= 18.752$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{pln}) \cdot e \cdot L / 2$$

$$= 105 / (2 \cdot 1200) \cdot 559 \cdot 711/2$$

$$= 8.742 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($18.8 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (18.8)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (18.8) / (8 \cdot 123) - (18.8)^3 / (8 \cdot 123^3))$$

$$= 149.1 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 3.61/149 + (8.74 \cdot 100/10666667) / 173$$

$$= 0.029$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	184.0	0.0	29.4	0.0	0.107E+04



Web	559.5	16.0	0.0	89.5	0.0	19.1
Totals	...	...	...	119.0	...	0.109E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/119$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200/2$$

$$= 100.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1050 - \cos(646/2)(16 + 0 + 0) - 0$$

$$= 388.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1085/119)}$$

$$= 30.204 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 388/30.2$$

$$= 12.846$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 105/(2 \cdot 1200)$$

$$= 0.044 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.044 \cdot 559 \cdot 388$$

$$= 9.541 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 9.54/108522$$

$$= 0.862 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($12.8 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.8)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (12.8) / (8 \cdot 123) - (12.8)^3 / (8 \cdot 123^3))$$

$$= 151.6 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 4.67/152 + (9.54 \cdot 100/1085) / 173$$

$$= 0.036$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	2.4000 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BlArea	1.6469 cm ²
Saddle Load Q0 (Weight)	Q0	11265.7 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	37.2 Kg
Maximum Transverse Force	Ft	165.9 Kg
Maximum Longitudinal Force	Fl	105.5 Kg
Saddle Bolted to Steel Foundation	No	



Shear Stress in a Single Bolt, Longitudinal Direction [τ_b, l]:

$$\begin{aligned} &= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\ &= 105 / (1.65 \cdot 2) \\ &= 3.1 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2. \end{aligned}$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b, t]:

$$\begin{aligned} &= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\ &= 166 / (1.65 \cdot 2) \\ &= 4.9 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2. \end{aligned}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [$Bltarear_l$]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [$Bltarears$]:

$$\begin{aligned} &= F_l / (\text{BoltShearAllowable} \cdot N_{bolts}) \\ &= 105 / (103 \cdot 2) \\ &= 0.0501 \text{ cm}^2 \end{aligned}$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [$Rmom$]:

$$\begin{aligned} &= B \cdot F_t + \text{Sum of X Moments} \\ &= 1050 \cdot 166 + 0 \\ &= 174.15 \text{ Kg-m.} \end{aligned}$$

Eccentricity (e):

$$\begin{aligned} &= Rmom / QO \\ &= 174 / 11266 \\ &= 15.46 \text{ mm.} < B_{plen}/6 \text{ --> No Uplift in Transverse direction} \end{aligned}$$

Bolt Area due to Transverse Load [$Bltareart$]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [$Bltarear$]:

$$\begin{aligned} &= \max[Bltarear_l, Bltarears, Bltareart] \\ &= \max[0, 0.05, 0] \\ &= 0.0501 \text{ cm}^2 \end{aligned}$$

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: T1-Tube Inlet
From: 20

Pressure for Reinforcement Calculations	P	15.520	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1220.00	mm.
Shell Finished (Minimum) Thickness	t	14.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		350.80	mm.
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		200.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	68.2680	mm.
Height of Beveled Transition	L'	35.2183	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		75.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: T1-Tube Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	200.000	mm.
Actual Thickness Used in Calculation	14.000	mm.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 15.5 \cdot 613 / (138 \cdot 1 - 0.6 \cdot 15.5)$
 $= 6.9412 \text{ mm.}$



Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 15.5 \cdot 103 / (138 \cdot 1 - 0.6 \cdot 15.5)$
 $= 1.1663 \text{ mm.}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 61 / 0.5773$
 $= 105.6551 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	412.0000	mm.
Parallel to Vessel Wall, opening length	d	206.0000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	14.299	NA	NA
Area in Shell	A1	8.361	NA	NA
Area in Nozzle Wall	A2	5.409	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	33.550	NA	NA
TOTAL AREA AVAILABLE	Atot	48.320	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (206 \cdot 6.94 \cdot 1 + 2 \cdot 72 \cdot 6.94 \cdot 1 \cdot (1 - 1))$
 $= 14.299 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 206 (1 \cdot 11 - 1 \cdot 6.94) - 2 \cdot 72$
 $(1 \cdot 11 - 1 \cdot 6.94) \cdot (1 - 1)$
 $= 8.361 \text{ cm}^2$



Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_{rn}) f_{r2}$$

$$= (2 \cdot 27.5) (11 - 1.17) 1$$

$$= 5.409 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot f_{r2} + (W_i - c_{an}/0.707)^2 \cdot f_{r2}$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(27.5, 300, 68.3)) \cdot (75 - 14) \cdot 1$$

$$= 33.550 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 4.1663 \text{ mm.}$
Wall Thickness per UG16(b),	$tr_{16b} = 5.5000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 9.9412 \text{ mm.}$
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 9.9412 \text{ mm.}$
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 5.5000 \text{ mm.}$
Wall Thickness per table UG-45	$tb_3 = 11.1026 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[11.1, \max(9.94, 5.5)]$$

$$= 9.9412 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(4.17, 9.94)$$

$$= 9.9412 \text{ mm.}$$

Available Nozzle Neck Thickness = 14.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 28.4,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 316.6 N./mm ²	Passed
Occasional	: 6.9,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 14.0,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

$$\text{Govrn. thk, } t_g = 14, tr = 1.17, c = 3 \text{ mm., } E^* = 1$$

$$\text{Thickness Ratio} = tr \cdot E^* / (t_g - c) = 0.11, \text{Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve A	2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

$$\text{Govrn. thk, } t_g = 14, tr = 6.94, c = 3 \text{ mm., } E^* = 1$$

$$\text{Thickness Ratio} = tr \cdot E^* / (t_g - c) = 0.63, \text{Temp. Reduction} = 21 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-39 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -39 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C



Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 15.52/51.00 = 0.304

Weld Size Calculations, Description: T1-Tube Inlet

Intermediate Calc. for nozzle/shell welds T_{min} 11.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•t_n•fr1•(E1•t-tr))•S_v)
= max(0, (14.3 - 8.36 + 2 • 72 • 1 •
(1 • 11 - 6.94))138)
= 16579.75 Kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(W_i-Can/.707)²•fr2)•S_v
= (39 + 1 - 0 • 1) • 138
= 56227.94 Kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • t_n • t • fr1)) • S_v
= (39 + 0 + 1 + (15.8)) • 138
= 78517.30 Kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2•t_n•t•fr1))•S
= (39 + 0 + 1 + 0 + (15.8)) • 138
= 78517.30 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [S_{onw}]:
= (π/2) • D_{lo} • W_o • 0.49 • S_{nw}
= (3.14/2.0) • 350 • 10 • 0.49 • 138
= 37908. Kgf

Shear, Nozzle Wall [S_{nw}]:
= (π • (D_{lr} + D_{lo})/4) • (Thk - Can) • 0.7 • S_n
= (3.14 • 139) • (75 - 3) • 0.7 • 138
= 309698. Kgf

Tension, Shell Groove Weld [T_{ngw}]:
= (π/2) • D_{lo} • (T - cas) • 0.74 • S_{ng}
= (3.14/2.0) • 350 • (14 - 3) • 0.74 • 138
= 62973. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (S_{onw} + S_{nw}) = (37908 + 309698) = 347605 Kgf

Failure Path [Path 2-2]:
= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})
= (37908 + 0 + 62973 + 0) = 100881 Kgf

Failure Path [Path 3-3]:
= (S_{onw} + T_{ngw} + S_{inw})

= (37908 + 62973 + 0) = 100881 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 347605 Kgf, must exceed W = 16579 Kgf or W1 = 56227 Kgf
 Path 2-2 = 100880 Kgf, must exceed W = 16579 Kgf or W2 = 78517 Kgf
 Path 3-3 = 100880 Kgf, must exceed W = 16579 Kgf or W3 = 78517 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 24.4 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 25.6414 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 339.6414 mm.

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 174.30 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 206.00 mm.

Input Echo, WRC297 Item 1, Description: T1-Tube Inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1220.000	mm.
Shell Thickness	14.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	200.000	mm.
Nozzle Thickness	75.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	15.52	bars
Radial Load	P	-1223.74	Kgf
Circumferential Shear	Vc	1223.74	Kgf
Longitudinal Shear	Vl	917.81	Kgf
Circumferential Moment	Mc	636.10	Kg-m.
Longitudinal Moment	Ml	654.45	Kg-m.
Torsional Moment	Mt	733.96	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (4.667) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

= NozzleOD + 2 • 1.65 • √(Rmean(t - ca))
 = 350.0 + 2 • 1.65 • √(618.5(14.0 - 3.0))



= 622.195 mm.

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1237.000 mm.
 Nozzle Outside Diameter (d) = 350.000 mm.
 Vessel Thickness used (T) = 11.000 mm.
 Nozzle Thickness used (t) = 72.000 mm.
 T / t = 0.153
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.000$$

Nr/P = 0.061
 Mr/P = 0.107
 M0/P = 0.032
 N0/P = 0.052
 MrD/Mc = 0.267
 NrDL/Mc = 0.100
 M0d/Mc = 0.080
 N0DL/Mc = 0.075
 MrD/Ml = 0.082
 NrDL/Ml = 0.073
 M0D/Ml = 0.024
 N0DL/Ml = 0.065

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-9	-9	9	9
Outplane Bending (ML)	-21	21	21	-21
Normal Pressure Stress	85	87	85	87
Outplane Stress Summary	78	85	141	62

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	63	-63	63	-63
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-11	-11	11	11
Inplane Bending (ML)	-74	74	74	-74
Inplane Pressure Stress	42	42	42	42
Inplane Stress Summary	27	48	197	-77

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	5	5	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Top	Al Top	Bu Bottom	Bl Bottom
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	Outside	Inside	Outside	Inside
Two * Max Shear Stress	78	86	197	139

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	-11	-11	11	11
Outplane Bending (Mc)	-70	70	70	-70
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	42	42	42	42
Outplane Stress Summary	-15	89	148	-30

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	63	-63	63	-63
Inplane Membrane (Mc)	-14	-14	14	14
Inplane Bending (Mc)	-236	236	236	-236
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	85	87	85	87
Inplane Stress Summary	-95	251	406	-191

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	1	1	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	95	251	406	192

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-9	-9	9	9
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	3	1	3
Hoop Stress Summary	-2	0	17	18

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to	Au Top	Al Top	Bu Bottom	Bl Bottom
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pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0
Axial Stress Summary	0	0	4	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	2	1	17	18

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-11	-11	11	11
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	3	1	3
Hoop Stress Summary	-3	-1	18	20

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-4	4	4	-4
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0
Axial Stress Summary	-3	2	8	-2

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Stress Intensities Outside Inside Outside Inside

Two * Max Shear Stress 4 4 18 22

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	85	87	85	87	85	87	85	87
Circ. Pl (SUS)	-4	-4	14	14	-8	-8	20	20
Circ. Q (SUS)	-3	3	39	-39	-173	173	299	-299
Long. Pm (SUS)	42	42	42	42	42	42	42	42
Long. Pl (SUS)	-5	-5	17	17	-6	-6	16	16
Long. Q (SUS)	-11	11	137	-137	-52	52	88	-88
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	85.0	87.0	85.0	87.0	85.0	87.0	85.0	87.0
Pm+Pl (SUS)	81.0	83.0	99.0	101.0	77.0	79.0	105.0	107.0
Pm+Pl+Q (Total)	78.3	86.4	196.1	140.1	96.0	252.0	404.1	192.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	87.00	138.00	Passed
Pm+Pl (SUS)	107.02	207.00	Passed
Pm+Pl+Q (TOTAL)	404.06	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1	3	1	3	1	3	1	3
Circ. Pl (SUS)	-4	-4	14	14	-6	-6	16	16
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	1	-1	1	-1	-3	3	5	-5
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1.0	3.0	1.0	3.0	1.0	3.0	1.0	3.0
Pm+Pl (SUS)	3.0	1.0	15.0	17.0	5.0	3.0	17.0	19.0
Pm+Pl+Q (Total)	3.0	2.0	15.0	17.0	5.0	5.0	17.0	23.0

Nozzle Stress Summation Comparison (N./mm²):



Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	3.00	138.00	Passed
Pm+Pl (SUS)	19.00	207.00	Passed
Pm+Pl+Q (TOTAL)	23.00	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: T2-Tube outlet

From: 20

Pressure for Reinforcement Calculations	P	15.400	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1220.00	mm.
Shell Finished (Minimum) Thickness	t	14.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		350.80	mm.
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		200.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	68.2680	mm.
Height of Beveled Transition	L'	35.2183	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		75.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.7000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: T2-Tube outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	200.000	mm.
Actual Thickness Used in Calculation	14.000	mm.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 15.4 \cdot 613 / (138 \cdot 1 - 0.6 \cdot 15.4)$
 $= 6.8874 \text{ mm.}$



Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 15.4 \cdot 103 / (138 \cdot 1 - 0.6 \cdot 15.4)$
 $= 1.1573 \text{ mm.}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 61 / 0.5773$
 $= 105.6551 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	412.0000	mm.
Parallel to Vessel Wall, opening length	d	206.0000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	14.188	NA	NA
Area in Shell	A1	8.472	NA	NA
Area in Nozzle Wall	A2	5.414	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	33.550	NA	NA
TOTAL AREA AVAILABLE	Atot	48.436	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (206 \cdot 6.89 \cdot 1 + 2 \cdot 72 \cdot 6.89 \cdot 1 \cdot (1 - 1))$
 $= 14.188 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 206(1 \cdot 11 - 1 \cdot 6.89) - 2 \cdot 72$
 $(1 \cdot 11 - 1 \cdot 6.89) \cdot (1 - 1)$
 $= 8.472 \text{ cm}^2$



Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_{rn}) f_{r2}$$

$$= (2 \cdot 27.5) (11 - 1.16) 1$$

$$= 5.414 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot f_{r2} + (W_i - c_{an}/0.707)^2 \cdot f_{r2}$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(27.5, 300, 68.3)) \cdot (75 - 14) \cdot 1$$

$$= 33.550 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 4.1573 \text{ mm.}$
Wall Thickness per UG16(b),	$tr_{16b} = 5.5000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 9.8874 \text{ mm.}$
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 9.8874 \text{ mm.}$
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 5.5000 \text{ mm.}$
Wall Thickness per table UG-45	$tb_3 = 11.1026 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[11.1, \max(9.89, 5.5)]$$

$$= 9.8874 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(4.16, 9.89)$$

$$= 9.8874 \text{ mm.}$$

Available Nozzle Neck Thickness = 14.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 28.0,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 317.0 N./mm ²	Passed
Occasional	: 6.8,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 14.0,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govern. thk, $t_g = 14$, $t_r = 1.16$, $c = 3 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govern. thk, $t_g = 14$, $t_r = 6.89$, $c = 3 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.63$, Temp. Reduction = 21 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-39 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -39 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C



Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 15.40/51.00 = 0.302

Weld Size Calculations, Description: T2-Tube outlet

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (14.2 - 8.47 + 2 • 72 • 1 •
(1 • 11 - 6.89))138)
= 16376.69 Kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (39 + 1 - 0 • 1) • 138
= 56234.93 Kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (39 + 0 + 1 + (15.8)) • 138
= 78524.31 Kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (39 + 0 + 1 + 0 + (15.8)) • 138
= 78524.31 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.14/2.0) • 350 • 10 • 0.49 • 138
= 37908. Kgf

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.14 • 139) • (75 - 3) • 0.7 • 138
= 309698. Kgf

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.14/2.0) • 350 • (14 - 3) • 0.74 • 138
= 62973. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (37908 + 309698) = 347605 Kgf

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (37908 + 0 + 62973 + 0) = 100881 Kgf

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)

= (37908 + 62973 + 0) = 100881 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 347605 Kgf, must exceed W = 16376 Kgf or W1 = 56234 Kgf
 Path 2-2 = 100880 Kgf, must exceed W = 16376 Kgf or W2 = 78524 Kgf
 Path 3-3 = 100880 Kgf, must exceed W = 16376 Kgf or W3 = 78524 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 24.4 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 25.6414 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 339.6414 mm.

Input Echo, WRC297 Item 1, Description: T2-Tube outlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1220.000	mm.
Shell Thickness	14.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	200.000	mm.
Nozzle Thickness	75.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUS tained loads.

Design Internal Pressure	Dp	15.40	bars
Radial Load	P	-1223.74	Kgf
Circumferential Shear	Vc	1223.74	Kgf
Longitudinal Shear	Vl	917.81	Kgf
Circumferential Moment	Mc	636.10	Kg-m.
Longitudinal Moment	Ml	636.10	Kg-m.
Torsional Moment	Mt	733.96	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (4.667) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 = NozzleOD + 2 • 1.65 • $\sqrt{R_{mean}(t - ca)}$
 = 350.0 + 2 • 1.65 • $\sqrt{618.5(14.0 - 3.0)}$
 = 622.195 mm.

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1237.000 mm.
 Nozzle Outside Diameter (d) = 350.000 mm.
 Vessel Thickness used (T) = 11.000 mm.
 Nozzle Thickness used (t) = 72.000 mm.
 T / t = 0.153
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.000$$

Nr/P = 0.061
 Mr/P = 0.107
 M0/P = 0.032
 N0/P = 0.052
 MrD/Mc = 0.267
 NrDL/Mc = 0.100
 M0d/Mc = 0.080
 N0DL/Mc = 0.075
 MrD/M1 = 0.082
 NrDL/M1 = 0.073
 M0D/M1 = 0.024
 N0DL/M1 = 0.065

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-9	-9	9	9
Outplane Bending (ML)	-20	20	20	-20
Normal Pressure Stress	85	86	85	86
Outplane Stress Summary	78	84	139	61

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	63	-63	63	-63
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-10	-10	10	10
Inplane Bending (ML)	-72	72	72	-72
Inplane Pressure Stress	42	42	42	42
Inplane Stress Summary	29	46	195	-76

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	5	5	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	79	85	195	138

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	-11	-11	11	11
Outplane Bending (Mc)	-70	70	70	-70
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	42	42	42	42
Outplane Stress Summary	-15	88	148	-30

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	63	-63	63	-63
Inplane Membrane (Mc)	-14	-14	14	14
Inplane Bending (Mc)	-236	236	236	-236
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	85	86	85	86
Inplane Stress Summary	-96	250	405	-192

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	1	1	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	96	250	405	192

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-9	-9	9	9
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	3	1	3
Hoop Stress Summary	-2	0	16	18

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1

Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	0	0	4	0
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	2	1	16	18

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-11	-11	11	11
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	3	1	3
Hoop Stress Summary	-3	-1	18	20

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-4	4	4	-4
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0
Axial Stress Summary	-3	2	8	-2

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	4	4	18	22

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	85	86	85	86	85	86	85	86
Circ. Pl (SUS)	-4	-4	14	14	-8	-8	20	20
Circ. Q (SUS)	-2	2	38	-38	-173	173	299	-299
Long. Pm (SUS)	42	42	42	42	42	42	42	42
Long. Pl (SUS)	-4	-4	16	16	-6	-6	16	16
Long. Q (SUS)	-9	9	135	-135	-52	52	88	-88
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	85.0	86.0	85.0	86.0	85.0	86.0	85.0	86.0
Pm+Pl (SUS)	81.0	82.0	99.0	100.0	77.0	78.0	105.0	106.0
Pm+Pl+Q (Total)	79.3	84.4	193.1	139.1	96.0	251.0	404.1	193.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	86.00	138.00	Passed
Pm+Pl (SUS)	106.02	207.00	Passed
Pm+Pl+Q (TOTAL)	404.06	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1	3	1	3	1	3	1	3
Circ. Pl (SUS)	-4	-4	14	14	-6	-6	16	16
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	1	-1	1	-1	-3	3	5	-5
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1.0	3.0	1.0	3.0	1.0	3.0	1.0	3.0
Pm+Pl (SUS)	3.0	1.0	15.0	17.0	5.0	3.0	17.0	19.0
Pm+Pl+Q (Total)	3.0	2.0	15.0	17.0	5.0	5.0	17.0	23.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	3.00	138.00	Passed

PV Elite 28 Licensee: SL User
FileName : H2 Storage aftercooler-AK
Nozzle Calcs.: T2-Tube outlet

Nozl: 10 3:41pm

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Pm+Pl (SUS)	19.00	207.00	Passed
Pm+Pl+Q (TOTAL)	23.00	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: S1-Shell inlet

From: 40

Pressure for Reinforcement Calculations	P	20.000	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1220.00	mm.
Shell Finished (Minimum) Thickness	t	20.0000	mm.
Shell Internal Corrosion Allowance	c	2.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1591.98	mm.
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		500.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	56.9504	mm.
Height of Beveled Transition	L'	36.9504	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm.
Corrosion Allowance	can	2.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: S1-Shell inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	500.000	mm.
Actual Thickness Used in Calculation	16.000	mm.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]



$$\begin{aligned}
 &= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= 20 \cdot 612 / (138 \cdot 1 - 0.6 \cdot 20) \\
 &= 8.9481 \text{ mm.}
 \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned}
 &= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= 20 \cdot 252 / (138 \cdot 1 - 0.6 \cdot 20) \\
 &= 3.6845 \text{ mm.}
 \end{aligned}$$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$\begin{aligned}
 &= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30) \\
 &= 64 / 0.5773 \\
 &= 110.8513 \text{ mm.}
 \end{aligned}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	1008.0000	mm.
Parallel to Vessel Wall, opening length	d	504.0000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	45.0000	mm.

Weld Strength Reduction Factor [fr1]:

$$\begin{aligned}
 &= \min(1, S_n / S_v) \\
 &= \min(1, 138 / 138) \\
 &= 1.000
 \end{aligned}$$

Weld Strength Reduction Factor [fr2]:

$$\begin{aligned}
 &= \min(1, S_n / S_v) \\
 &= \min(1, 138 / 138) \\
 &= 1.000
 \end{aligned}$$

Weld Strength Reduction Factor [fr3]:

$$\begin{aligned}
 &= \min(fr2, fr4) \\
 &= \min(1, 1) \\
 &= 1.000
 \end{aligned}$$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	45.098	NA	NA
Area in Shell	A1	45.622	NA	NA
Area in Nozzle Wall	A2	9.284	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	57.600	NA	NA
TOTAL AREA AVAILABLE	Atot	113.506	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (504 \cdot 8.95 \cdot 1 + 2 \cdot 78 \cdot 8.95 \cdot 1 \cdot (1 - 1)) \\
 &= 45.098 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$



$$= 504 (1 \cdot 18 - 1 \cdot 8.95) - 2 \cdot 78$$

$$(1 \cdot 18 - 1 \cdot 8.95) \cdot (1 - 1)$$

$$= 45.622 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_{rn}) f_{r2}$$

$$= (2 \cdot 45) (14 - 3.68) 1$$

$$= 9.284 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot f_{r2} + (W_i - c_n / 0.707)^2 \cdot f_{r2}$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(45, 200, 57)) \cdot (80 - 16) \cdot 1$$

$$= 57.600 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 5.6845 \text{ mm.}$
Wall Thickness per UG16(b),	$t_{r16b} = 3.5000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$t_{rb1} = 10.9481 \text{ mm.}$
Wall Thickness	$t_{b1} = \max(t_{rb1}, t_{r16b}) = 10.9481 \text{ mm.}$
Wall Thickness	$t_{b2} = \max(t_{rb2}, t_{r16b}) = 3.5000 \text{ mm.}$
Wall Thickness per table UG-45	$t_{b3} = 10.3312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[t_{b3}, \max(t_{b1}, t_{b2})]$$

$$= \min[10.3, \max(10.9, 3.5)]$$

$$= 10.3312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(5.68, 10.3)$$

$$= 10.3312 \text{ mm.}$$

Available Nozzle Neck Thickness = 16.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 16$, $t_r = 3.68$, $c = 2 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.26$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	6 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, $t_g = 20$, $t_r = 8.95$, $c = 2 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.5$, Temp. Reduction = 33 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-40 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -40 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $20.00 / 51.00 = 0.392$

Weld Size Calculations, Description: S1-Shell inlet



Intermediate Calc. for nozzle/shell Welds Tmin 18.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (45.1 - 45.6 + 2 \cdot 78 \cdot 1 \cdot (1 \cdot 18 - 8.95)) \cdot 138)$
 $= 19133.75 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (66.9 + 1 - 0 \cdot 1) \cdot 138$
 $= 95523.38 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (66.9 + 0 + 1 + (28.1)) \cdot 138$
 $= 135036.34 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (66.9 + 0 + 1 + 0 + (28.1)) \cdot 138$
 $= 135036.34 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 660 \cdot 10 \cdot 0.49 \cdot 138$
 $= 71483. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 291) \cdot (80 - 2) \cdot 0.7 \cdot 138$
 $= 702390. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 660 \cdot (14 - 0) \cdot 0.74 \cdot 138$
 $= 151135. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (71483 + 702390) = 773872 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (71483 + 0 + 151135 + 0) = 222618 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (71483 + 151135 + 0) = 222618 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 773872 Kgf, must exceed W = 19133 Kgf	or W1 = 95523 Kgf
Path 2-2 = 222618 Kgf, must exceed W = 19133 Kgf	or W2 = 135036 Kgf
Path 3-3 = 222618 Kgf, must exceed W = 19133 Kgf	or W3 = 135036 Kgf



Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 34.4 bars

The Drop for this Nozzle is : 96.9698 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 316.9698 mm.

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Input, Nozzle Desc: S2-Shell outlet

From: 60

Pressure for Reinforcement Calculations	P	20.138	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1220.00	mm.
Shell Finished (Minimum) Thickness	t	20.0000	mm.
Shell Internal Corrosion Allowance	c	2.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		6091.97	mm.
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		500.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	56.9504	mm.
Height of Beveled Transition	L'	36.9504	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm.
Corrosion Allowance	can	2.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: S2-Shell outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	500.000	mm.
Actual Thickness Used in Calculation	16.000	mm.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]



$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 20.1 \cdot 612 / (138 \cdot 1 - 0.6 \cdot 20.1)$$

$$= 9.0101 \text{ mm.}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 20.1 \cdot 252 / (138 \cdot 1 - 0.6 \cdot 20.1)$$

$$= 3.7101 \text{ mm.}$$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

$$= 64 / 0.5773$$

$$= 110.8513 \text{ mm.}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	1008.0000	mm.
Parallel to Vessel Wall, opening length	d	504.0000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	45.0000	mm.

Weld Strength Reduction Factor [fr1]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 138 / 138)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr2]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 138 / 138)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(1, 1)$$

$$= 1.000$$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	45.411	NA	NA
Area in Shell	A1	45.309	NA	NA
Area in Nozzle Wall	A2	9.261	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	57.600	NA	NA
TOTAL AREA AVAILABLE	Atot	113.170	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (504 \cdot 9.01 \cdot 1 + 2 \cdot 78 \cdot 9.01 \cdot 1 \cdot (1 - 1))$$

$$= 45.411 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$



$$= 504 (1 \cdot 18 - 1 \cdot 9.01) - 2 \cdot 78 \\ (1 \cdot 18 - 1 \cdot 9.01) \cdot (1 - 1) \\ = 45.309 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np})(t_n - t_{rn})fr_2 \\ = (2 \cdot 45)(14 - 3.71)1 \\ = 9.261 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2 \\ = 10^2 \cdot 1 + (0)^2 \cdot 1 \\ = 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot fr_2 \\ = (2 \cdot \min(45, 200, 57)) \cdot (80 - 16) \cdot 1 \\ = 57.600 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$ta = 5.7101 \text{ mm.}$
Wall Thickness per UG16(b),	$tr16b = 3.5000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$trb1 = 11.0101 \text{ mm.}$
Wall Thickness	$tb1 = \max(trb1, tr16b) = 11.0101 \text{ mm.}$
Wall Thickness	$tb2 = \max(trb2, tr16b) = 3.5000 \text{ mm.}$
Wall Thickness per table UG-45	$tb3 = 10.3312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)] \\ = \min[10.3, \max(11, 3.5)] \\ = 10.3312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb) \\ = \max(5.71, 10.3) \\ = 10.3312 \text{ mm.}$$

Available Nozzle Neck Thickness = 16.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 16$, $t_r = 3.71$, $c = 2 \text{ mm.}$, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.27$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A	$6 \text{ }^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	$-104 \text{ }^\circ\text{C}$

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, $t_g = 20$, $t_r = 9.01$, $c = 2 \text{ mm.}$, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.5$, Temp. Reduction = $32 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B	$-7 \text{ }^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	$-40 \text{ }^\circ\text{C}$
Min Metal Temp. w/o impact per UG-20(f)	$-29 \text{ }^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : $-40 \text{ }^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	$-18 \text{ }^\circ\text{C}$
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	$-48 \text{ }^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $20.14/51.00 = 0.395$

Weld Size Calculations, Description: S2-Shell outlet



Intermediate Calc. for nozzle/shell Welds Tmin 18.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (45.4 - 45.3 + 2 • 78 • 1 •
(1 • 18 - 9.01))138)
= 19877.93 Kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (66.9 + 1 - 0 • 1) • 138
= 95491.01 Kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (66.9 + 0 + 1 + (28.1)) • 138
= 135003.97 Kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (66.9 + 0 + 1 + 0 + (28.1)) • 138
= 135003.97 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.14/2.0) • 660 • 10 • 0.49 • 138
= 71483. Kgf

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.14 • 291) • (80 - 2) • 0.7 • 138
= 702390. Kgf

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.14/2.0) • 660 • (14 - 0) • 0.74 • 138
= 151135. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (71483 + 702390) = 773872 Kgf

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (71483 + 0 + 151135 + 0) = 222618 Kgf

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (71483 + 151135 + 0) = 222618 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 773872 Kgf, must exceed W = 19877 Kgf	or W1 = 95491 Kgf
Path 2-2 = 222618 Kgf, must exceed W = 19877 Kgf	or W2 = 135003 Kgf
Path 3-3 = 222618 Kgf, must exceed W = 19877 Kgf	or W3 = 135003 Kgf



Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 34.3 bars

The Drop for this Nozzle is : 96.9698 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 316.9698 mm.

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: main Shell-1

Shell Maximum Design Pressure	Psd,max	20.00 bars
Shell Maximum Operating Pressure	Psox,max	20.00 bars
Shell Minimum Operating Pressure	Psox,min	0.00 bars
Shell Thickness	ts	20.0000 mm.
Shell Internal Corrosion Allowance	cas	2.0000 mm.
Shell External Corrosion Allowance	casext	0.0000 mm.
Inside Diameter of Shell	Ds	1220.000 mm.
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	85.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	15.40 bars
Channel Maximum Operating Pressure	Ptox,max	15.40 bars
Channel Minimum Operating Pressure	Ptox,min	0.00 bars
Channel Thickness	tc	14.0000 mm.
Channel Internal Corrosion Allowance	cac	3.0000 mm.
Channel External Corrosion Allowance	cacext	0.0000 mm.
Inside Diameter of Channel	Dc	1220.000 mm.
Channel Design Temperature	TEMPC	85.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	722
Tube Wall Thickness	et	2.1100 mm.
Tube Outside Diameter	D	25.4000 mm.
Total Straight Tube Length	Lt	6069.00 mm.
Straight Tube Length (bet. inner tubsht faces) L		5889.00 mm.
Design Temperature of the Tubes		85.00 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		92.40 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	165.14 N./mm ²
Tube Pitch (Center to Center Spacing)	P	38.1000 mm.
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	2.1100 mm.
Groove Weld Leg	ag	2.1100 mm.
Tube-Tubesheet Joint Weld Type	Full Strength	
Method for Tube-Tubesheet Jt. Allowable	UW-20.3	
Is the Tube Inset	No	

Radius to Outermost Tube Hole Center	ro	590.520 mm.
Largest Center-to-Center Tube Distance	Ul	0.0000 mm.
Length of Expanded Portion of Tube	ltx	0.0000 mm.
Tube-side pass partition groove depth	hg	6.0000 mm.
Area of Tube Layout	Ap	9184.1 cm ²
Perimeter of Tube Layout	Cp	3649.0 mm.



Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf B

Tubesheet Design Metal Temperature T 85.00 °C
 Tubesheet Material SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03506
 Tubesheet Allowable Stress at Temperature S 138.00 N./mm²
 Tubesheet Allowable Stress at Ambient Tt 138.00 N./mm²
 Thickness of Tubesheet h 90.0000 mm.
 Tubesheet Corr. Allowance (Shell side) Cats 2.0000 mm.
 Tubesheet Corr. Allowance (Channel side) Catc 3.0000 mm.
 Tubesheet Outside Diameter A 1430.000 mm.

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped? No
 Area of the Untubed Lanes AL 0.0 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration l 823.000 mm.
 Tube End condition corresponding to Span (l) k 0.80
 Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes
 Mean Shell Metal Temp. along Shell len. Tsm 0.00 °C
 Mean Tube Metal Temp. along Tube length Ttm 0.00 °C
 Junction Stress Reduction option Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side Channel
 Flange Outside Diameter A 1430.000 mm.
 Flange Inside Diameter B 1220.000 mm.
 Flange Face Outside Diameter Fod 1301.000 mm.
 Flange Face Inside Diameter Fid 1220.000 mm.
 Gasket Outside Diameter Go 1290.000 mm.
 Gasket Inside Diameter Gi 1252.000 mm.
 Small end Hub thk. g0 14.0000 mm.
 Large end Hub thk. g1 20.0000 mm.
 Gasket Factor, m 3.00
 Gasket Design Seating Stress y 68.95 N./mm²
 Flange Facing Sketch Code Sketch 1a
 Gasket Thickness tg 3.1750 mm.
 Full face Gasket Flange Option Program Selects

Bolting Information:

Diameter of Bolt Circle C 1360.000 mm.
 Nominal Bolt Diameter dB 30.0000 mm.
 Thread Series TEMA Metric Thread
 Number of Bolts n 60
 Bolt Material SA-193 B7
 Bolt Allowable Stress At Temperature Sb 172.00 N./mm²
 Bolt Allowable Stress At Ambient Sa 172.00 N./mm²
 Weld between Flange and Shell/Channel 0.0000 mm.

Note: The full bolt load Sa*Ab, was used for all load cases.

Tubesheet Integral with Shell
 Tubesheet Extended as Flange Yes
 Thickness of Extended Portion of Tubesheet Tf 54.0000 mm.
 Is Bolt Load Transferred to the Tubesheet Yes

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No



ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	85.0 °C	0.19904E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	85.0 °C	0.19904E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	85.0 °C	0.19904E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	85.0 °C	0.19904E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	85.0 °C	0.19904E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (15.4 \cdot (25.4/2 - 0)) / (92.4 \cdot 1 + 0.4 \cdot 15.4)$
 $= 0.2103 + 0.0000 = 0.2103 \text{ mm.}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 85.00 °C
 Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	25.40	5889.00	12.04	50.0000	0.0075908	95.82

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 95.8) / (3 \cdot 12) = 106 \text{ bars}$

Results for Required Thickness: No Conversion

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.6010	25.40	5889.00	42.26	50.0000	0.0006159	61.57

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 61.6) / (3 \cdot 42.3) = 19.4 \text{ bars}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.6010 mm.
Allowable Internal Pressure at Corroded thickness	164.43 bars
Required Internal Design Pressure	15.40 bars
Allowable External Pressure at Corroded thickness	106.12 bars
Required External Design Pressure	20.00 bars



Required Thickness due to Shell Side pressure 0.6010 mm.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Goc - Gic) / 2$	19.000 mm.
Basic Gasket Width,	$b0 = N / 2$	9.500 mm.
Effective Gasket Width,	$b = \text{sqrt}(b0) * 2.5$	7.764 mm.
Gasket Reaction Diameter,	$G = Go - 2.0 * b$	1274.472 mm.

Results for ASME Fixed Tubesheet Calculations for Configuration b:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [p]:
 $= ltx / h$ (modified for corrosion if present)
 $= 0/90$
 $= 0.0000$ (must be $0 \leq p \leq 1$)

Effective tube hole diameter [d*]:
 $= \max(dt - 2 * tt(EtT/E) * (StT/S) * (p); dt - 2 * tt)$
 $= \max(25.4 - 2 * 2.11 (199042256/199042256) * (92.4/138) * (0); 25.4 - 2 * 2.11)$
 $= 25.4000 \text{ mm.}$

Equivalent outer tube limit circle diameter [Do]:
 $= 2 * ro + dt$
 $= 2 * 591 + 25.4$
 $= 1206.440 \text{ mm.}$

Determine the basic ligament efficiency for shear [μ]:
 $= (p - dt) / p$
 $= (38.1 - 25.4) / 38.1$
 $= 0.333$

Equivalent outer tube limit radius [ao]:
 $= Do / 2$
 $= 1206/2$
 $= 603.2200 \text{ mm.}$

Effective tube pitch [p*]:
 $= p / \sqrt{ (1 - 4 * \min(AL * CNVfactor; 4 * Do * p) / (\pi * Do^2)) }$
 $= 38.1 / \sqrt{ (1 - 4 * \min(0 * 100 ; 4 * 1206 * 38.1) / (\pi * 1206^2)) }$
 $= 38.1000 \text{ mm.}$

Effective ligament efficiency for bending [μ^*]:
 $= (p^* - d^*) / p^*$
 $= (38.1 - 25.4) / 38.1$
 $= 0.3333$

Ratio [ps]:
 $= as / ao$
 $= 610/603$
 $= 1.0112$

Ratio [pc]:
 $= ac / ao$
 $= 637/603$
 $= 1.0564$

Compute parameter [xt]:
 $= 1 - Nt((dt - 2 * tt) / (2 * ao))^2$



$$= 1 - 722 \left((25.4 - 2 \cdot 2.11) / (2 \cdot 603) \right)^2$$

$$= 0.7775$$

Determine parameter [xs]:

$$= 1 - Nt \left(dt / (2 \cdot ao) \right)^2$$

$$= 1 - 722 \left(25.4 / (2 \cdot 603) \right)^2$$

$$= 0.6800$$

Determine the value [h'g]:

$$= \max((hg - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((6 - 0); 0)$$

$$= 6.000 \text{ mm.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot ts \cdot (Ds + ts) Es / L$$

$$= \pi \cdot 20 \cdot (1220 + 20) 199042256 / 5889$$

$$= 268535.5938 \text{ Kg/mm.}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot tt \cdot (Dt - tt) Et / L$$

$$= \pi \cdot 2.11 \cdot (25.4 - 2.11) 199042256 / 5889$$

$$= 532.1108 \text{ Kg/mm.}$$

Stiffness Factor [Ks,t]:

$$= Ks / (Nt \cdot Kt)$$

$$= 268536 / (722 \cdot 532)$$

$$= 0.69898$$

Rigidity Ratio [J]:

$$= 1 / (1 + Ks / Kj)$$

$$= 1 / (1 + 268536 / 0)$$

$$= 1.0000 \text{ (= 1 if No Exp. Jt.)}$$

Shell coefficient [βs]:

$$= \left((12(1 - vs^2))^{1/4} \right) / \left((Ds + ts) ts \right)$$

$$= \left((12(1 - 0.3^2))^{1/4} \right) / \left((1220 + 20) 20 \right)$$

$$= 0.0115 \text{ 1/mm.}$$

Shell coefficient [ks]:

$$= \beta s \cdot Es \cdot ts^3 / (6(1 - vs^2))$$

$$= 0.012 \cdot 199042256 \cdot 20^3 / (6(1 - 0.3^2))$$

$$= 33664512.0000 \text{ bars} \cdot \text{mm.}^2$$

Shell coefficient [λs]:

$$= 6(Ds + ts) \cdot ks / h^3 \cdot (1 + h \cdot \beta s + (h \cdot \beta s)^2 / 2)$$

$$= 6(1220 + 20) \cdot 33664512 / 90^3 \cdot (1 + 90 \cdot 0.012 + (90 \cdot 0.012)^2 / 2)$$

$$= 885917.4375 \text{ bars}$$

Shell coefficient [δs]:

$$= Ds^2 / (4 \cdot Es \cdot ts) \cdot (1 - Ds \cdot vs / (2(Ds + ts)))$$

$$= 1220^2 / (4 \cdot 199042256 \cdot 20) \cdot (1 - 1220 \cdot 0.3 / (2(1220 + 20)))$$

$$= 0.79673 \text{E-01 mm./N./mm}^2$$

Intermediate parameters for Tubesheet Gasketed on the Channel Side:

βC, kc, λC, δC are all = 0

ASME Section VIII-2 Part 4.18.8.4 Step 3:

h/p = 2.362205 ; μ* = 0.333333 ; E* = 64706924 KPa.

E*/E = 0.325091 ; ν* = 0.334167

Note: As h/p (2.362) is > 2, data values for h/p = 2 were used.

Tube Bundle Stiffness Factor [Xa]:

$$= \left((24(1 - \nu^{*2}) Nt \cdot Et \cdot tt(dt - tt) ao^2) / (E* \cdot L \cdot H^3) \right)^{1/4}$$

$$= \left((24(1 - 0.33^2) 722 \cdot 199042256 \cdot 2.11 \right. \\ \left. (25.4 - 2.11) 603^2) / (64706924 \cdot 5889 \cdot 90^3) \right)^{1/4}$$

$$= 3.7475$$



Values from Table 4.18.3

$$\begin{aligned} Z_d &= 0.029215 ; Z_v = 0.072525 ; Z_m = 0.399799 \\ Z_a &= 0.500376E+01 ; Z_w = 0.072525 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$\begin{aligned} &= A / D_o \\ &= 1430/1206 \\ &= 1.1853 \end{aligned}$$

Compute coefficient [F]:

$$\begin{aligned} &= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\ &= (1 - 0.33) / 64706924 \cdot (885917 + 0 + 199042256 \cdot \ln(1.19)) \\ &= 1.2598 \end{aligned}$$

Compute parameter [ϕ]:

$$\begin{aligned} &= (1 + v^*) F \\ &= (1 + 0.33) 1.26 \\ &= 1.6808 \end{aligned}$$

Compute parameter [Q1]:

$$\begin{aligned} &= (\psi_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m) \\ &= (1.01 - 1 - 1.68 \cdot 0.073) / (1 + 1.68 \cdot 0.4) \\ &= -0.066185 \end{aligned}$$

Compute Parameter [Qz1]:

$$\begin{aligned} &= (Z_d + Q1 \cdot Z_w) / 2 \cdot X_a^4 \\ &= (0.029 + -0.066 \cdot 0.073) / 2 \cdot 3.75^4 \\ &= 2.4076 \end{aligned}$$

Compute parameter [Qz2]:

$$\begin{aligned} &= (Z_v + Q1 \cdot Z_m) / 2 \cdot X_a^4 \\ &= (0.073 + -0.066 \cdot 0.4) / 2 \cdot 3.75^4 \\ &= 4.5425 \end{aligned}$$

Compute parameter [U]:

$$\begin{aligned} &= (Z_w + (\psi_s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m) \\ &= (0.073 + (1.01 - 1) 0.4) 3.75^4 / (1 + 1.68 \cdot 0.4) \\ &= 9.0849 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γ_b]:

$$\begin{aligned} &= (G_c - C) / D_o \text{ (config b)} \\ &= (1274 - 1360) / 1206 = -0.071 \end{aligned}$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm. (For Pressure only cases)}$$

Calculate parameter [ω_s]:

$$\begin{aligned} &= \psi_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s) \\ &= 1.01 \cdot 33664512 \cdot 0.012 \cdot 0.079673253 (1 + 90 \cdot 0.012) \\ &= 6383.9458 \text{ mm.}^2 \end{aligned}$$

Calculate parameter [$\omega \cdot s$]:

$$\begin{aligned} &= a o^2 (\psi_s^2 - 1) \cdot (\psi_s - 1) / 4 - \omega_s \\ &= 603^2 (1.01^2 - 1) \cdot (1.01 - 1) / 4 - 6384 \\ &= -6360.8320 \text{ mm.}^2 \end{aligned}$$

Parameter [ω_c]:

$$\begin{aligned} &= \rho_c \cdot k_c [\delta C (C_m \Theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \Theta + h \cdot C_q \Theta \cdot h \cdot \beta C / 2)] \\ &= 1.06 \cdot 0 [0.000 (0 + 0 \cdot 0 \cdot 90) - 0.000 (0/90 + 0 \cdot 0/2)] \\ &= 0.0000 \text{ mm.}^2 \end{aligned}$$

Parameter [$\omega \cdot c$]:



$$\begin{aligned}
 &= ao^2 [(pc^2 + 1) \cdot (pc - 1)/4 - (ps - 1)/2] - wc \\
 &= 603^2 [(1.06^2 + 1) \cdot (1.06 - 1)/4 - (1.01 - 1)/2] - 0 \\
 &= 8809.4990 \text{ mm}^2
 \end{aligned}$$

Pressure [P*S]:
 = 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [P*C]:
 = 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$\begin{aligned}
 &= Ps \{ xs + 2(1 - xs)vt + [2/Kst \cdot (Ds/Do)^2] vs - \\
 &\quad [(ps^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)] [Dj^2 - (Ds)^2 / Do^2] \} \\
 &= 20 \{ 0.68 + 2(1 - 0.68)0.3 + [2/0.7 \cdot \\
 &\quad (1220/1206)^2]0.3 - [(1.01^2 - 1)/(1 \cdot 0.7)] - \\
 &\quad [(1 - 1)/(2 \cdot 1 \cdot 0.7)] [0^2 - (1220)^2 / 1206^2] \} \\
 &= 34.3490 \text{ bars}
 \end{aligned}$$

Pressure [P't]:

$$\begin{aligned}
 &= [xt + 2(1 - xt)vt + 1/(J \cdot Kst)] Pt \\
 &= [0.78 + 2(1 - 0.78)0.3 + 1/(1 \cdot 0.7)] 15.4 \\
 &= 36.0615 \text{ bars}
 \end{aligned}$$

Pressure [Pγ]:

$$\begin{aligned}
 &= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2) \\
 &= 722 \cdot 532 \cdot 0 / (3.14 \cdot 603^2) \\
 &= 0.0000 \text{ bars}
 \end{aligned}$$

Pressure [Pw]:

$$\begin{aligned}
 &= -\gamma b \cdot U \cdot W* / (2 \cdot \pi \cdot ao^2) \\
 &= -0.071 \cdot 9.08 \cdot 529274 / (2 \cdot \pi \cdot 603^2) \\
 &= 14.6210 \text{ bars}
 \end{aligned}$$

Calculate the pressure [Prim]:

$$\begin{aligned}
 &= -(U/ao^2) \cdot (\omega*s \cdot Ps - \omega*c \cdot Pt) \\
 &= -(9.08/603^2) \cdot (-6361 \cdot 20 - 8809 \cdot 15.4) \\
 &= 6.5635 \text{ bars}
 \end{aligned}$$

Calculate the pressure [Pw]:

$$\begin{aligned}
 &= (U/ao^2) \cdot (\omega*s \cdot P*s - \omega*c \cdot P*c) \\
 &= (9.08/603^2) \cdot (6384 \cdot 0 - 0 \cdot 0) \\
 &= 0.0000 \text{ bars}
 \end{aligned}$$

Determine the effective pressure [Pe]:

$$\begin{aligned}
 &= J \cdot Kst / (1 + J \cdot Kst(Qz1 + (ps - 1)Qz2)) \cdot \\
 &\quad (P's - P't + P\gamma + Pw + Prim) \\
 &= 0.1000E+01 \cdot 0.7 / (1 + 1 \cdot 0.7(2.41 + (1.01 - 1)4.54)) \cdot (34.3 - 36.1 + 0 + 0 + 14.6 + 6.56) \\
 &= 5.0066 \text{ bars}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$\begin{aligned}
 &= [((\omega*s \cdot Ps - \omega*c \cdot Pt) - (\omega*s \cdot P*s - \omega*c \cdot P*c)) + \\
 &\quad W* \cdot \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Zm) \\
 &= [((-6361 \cdot 20 - 8809 \cdot 15.4) - \\
 &\quad (6384 \cdot 0 - 0 \cdot 0)) \cdot 0.01 + \\
 &\quad 529274 \cdot -0.071 / (2 \cdot \pi)] / (1 + 1.68 \cdot 0.4) \\
 &= -5175.0576 \text{ Kg}
 \end{aligned}$$

Calculate factor [Q3]:

$$\begin{aligned}
 &= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2) \\
 &= -0.066 + 2 \cdot -5175 / (5.01 \cdot 603^2) \\
 &= -0.62332
 \end{aligned}$$

Fm Value from Table 4.18.3 = 0.311658

Tubesheet Bending Stress - original thickness [σ]:



$$\begin{aligned}
 &= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (H - h'g))^2 \cdot P_e \\
 &= (1.5 \cdot 0.31 / 0.33) \cdot (2 \cdot 603 / (90 - 6))^2 \cdot 5.01 \\
 &= 144.8467 \text{ N./mm}^2
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ allowed]:
 $= 1.5 \cdot S = 1.5 \cdot 138 = 207 \text{ N./mm}^2$

Tubesheet Bending Stress - final thickness [σ_f]:
 $= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$
 $= (1.5 \cdot 0.25 / 0.33) \cdot (2 \cdot 603 / (74.2 - 6))^2 \cdot 5.81$
 $= 206.9935 \text{ N./mm}^2$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:
 $= h + C_{ats} + C_{atc}$
 $= 74.2 + 0 + 0$
 $= 74.1940 \text{ mm.}$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ_{limit}]:
 $= (2 \cdot \mu \cdot h / a_o) \cdot \min(0.8S; 0.533S_y)$
 $= (2 \cdot 0.33 \cdot 90 / 603) \cdot 110$
 $= 10.9809 \text{ N./mm}^2$

Shear Stress is not required to be computed; [P_e] $\leq$ τ_{limit}

Tubesheet Shear Stress is probably low, use the following req. thk:

Tubesheet thickness (Incl. Corr.) = 5.3057 mm.
 Tubesheet Shear Stress = 31.1248 N./mm²

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd]:
 $= \max(H_{reqB}; H_{reqS})$
 $= \max(74.2; 5.31)$
 $= 74.1940 \text{ mm.}$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:
 $F_{tmin} = -1.58$, $F_{tmax} = 1.95$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:
 $= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s)$
 $= ((20 \cdot 0.68 - 15.4 \cdot 0.78) - (5.01) \cdot -1.58) /$
 $(0.78 - 0.68)$
 $= 9.7650 \text{ N./mm}^2$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:
 $= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s)$
 $= ((20 \cdot 0.68 - 15.4 \cdot 0.78) - (5.01) \cdot 1.95) /$
 $(0.78 - 0.68)$
 $= -8.3637 \text{ N./mm}^2$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:
 $= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 9.77 \text{ N./mm}^2$

Allowable Tube Stress [$\sigma_{t,A}$]:
 $= S_{ot} = 92.3983 \text{ N./mm}^2$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:
 $= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25)$
 $= \max((3.25 - 0.25(0.029 + -0.62 \cdot 0.073)3.75^4); 1.25)$
 $= 2.0000 \text{ (Should be } \leq 2)$

Determine the factor [rt]:
 $= (\sqrt{dt^2 + (dt - 2 \cdot tt)^2}) / 4$
 $= (\sqrt{25.4^2 + (25.4 - 2 \cdot 2.11)^2}) / 4$
 $= 8.2680 \text{ mm.}$



Determine the factor [Ct]:

$$\begin{aligned} &= \sqrt{(2 \cdot \pi^2 \cdot Et/Syt)} \\ &= \sqrt{(2 \cdot \pi^2 \cdot 199042256/165)} \\ &= 154.2491 \end{aligned}$$

Determine the factor [Ft]:

$$\begin{aligned} &= k \cdot L/r \\ &= 0.8 \cdot 823/8.27 \\ &= 79.6325 \end{aligned}$$

Buckling Allowable Stress [Stb]:

$$\begin{aligned} &= (Sy/tFs) \cdot (1 - Ft/(2 \cdot Ct)) \\ &= (165/2) \cdot (1 - 79.6/(2 \cdot 154)) \\ &= 61.257 \text{ N/mm}^2 \quad (\text{Never greater than } Sot) \end{aligned}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$\begin{aligned} &= \sigma_{t,max} \cdot \text{Tube Area} \\ &= 9.77 \cdot 1.54 \\ &= 153.73 \text{ Kg} \end{aligned}$$

Tube Metal Area [At]:

$$\begin{aligned} &= \pi (do - t) t \\ &= \pi (25.4 - 2.11) 2.11 \\ &= 1.544 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$\begin{aligned} &= \pi \cdot t (do - t) S \\ &= \pi \cdot 2.11 (25.4 - 2.11) 92.4 \\ &= 1454.579 \text{ Kg} \end{aligned}$$

Groove Weld Strength [Fg]:

$$\begin{aligned} &= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \quad (\text{but not } > Ft) \\ &= 0.85 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 92.4 \\ &= 1423.4542 \text{ Kg} \end{aligned}$$

Weld Strength Factor [fw]:

$$\begin{aligned} &= S / Sw \quad \text{where } Sw = \min(S \text{ and } St) \\ &= 92.4/92.4 \\ &= 1.000 \end{aligned}$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$\begin{aligned} &= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \quad (\text{but not } > Ft) \\ &= 0.55 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 92.4 \\ &= 921.0587 \text{ Kg} \end{aligned}$$

Minimum Length of Weld [ar]:

$$\begin{aligned} &= 2 \left(\sqrt{(0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) fw \cdot fd} - 0.75 \cdot do \right) \\ &= 2 \left(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 2.11 \cdot (25.4 - 2.11) 1 \cdot 1} - 0.75 \cdot 25.4 \right) \\ &= 2.667 \text{ mm.} \end{aligned}$$

$$\text{Minimum Required Fillet Weld Leg [afr]:} \quad 1.3334 \text{ mm.}$$

$$\text{Minimum Required Groove Weld Leg [agr]:} \quad 1.3334 \text{ mm.}$$

Tube-Tubesheet Jt allowable, 1455 is $\geq$ tube strength 1455 Kg

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:



$$= S_s \cdot E_{sw} = 138 \cdot 1 = 138 \text{ N./mm}^2$$

Axial Membrane Stress in Shell $[\sigma_s, m]$:

$$= a o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1) (P_s - P_t)] + a s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 603^2 / ((1220 + 20) 20) \cdot [5.01 + (1.01^2 - 1) (20 - 15.4)] + 610^2 \cdot 15.4 / ((1220 + 20) 20)$$

$$= 30.6064 \text{ N./mm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 20 mm., the minimum Shell length adjacent to the tubesheet should be 281.169 mm.

Shell Membrane Stress due to Joint Interaction $[\sigma_s, m]$:

$$= a o^2 / ((D_s + t_s) t_s) [P_e + (p_s^2 - 1) (P_s - P_t)] + a s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 603^2 / ((1220 + 20) 20) [5.01 + (1.01^2 - 1) (20 - 15.4)] + 610^2 \cdot 15.4 / ((1220 + 20) 20)$$

$$= 30.6064 \text{ N./mm}^2$$

Shell Bending Stress due to Joint Interaction $[\sigma_s, b]$:

$$= 6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s \cdot P_s + a s^2 \cdot P \cdot s / (E_s \cdot t_s)] + 6 (1 - \nu^2) / (E^*) (a o / h)^3 (1 + h \cdot \beta_s / 2) [P_e (Z_v + Z_m \cdot Q_1) + 2 / a o^2 \cdot Z_m \cdot Q_2] \}$$

$$= 6 \cdot 33664512 / 20^2 \{ 0.012 [0.079673253 \cdot 20 + 610^2 \cdot 0 / (3.98E+09)] + 6 (1 - 0.33^2) / (64706924) (603/90)^3 (1 + 90 \cdot 0.012/2) [5.01 (0.073 + 0.4 \cdot -0.066) + 2 / 603^2 \cdot 0.4 \cdot -5175] \}$$

$$= -75.4385 \text{ N./mm}^2$$

Shell Stress Summation vs. Allowable
 $\therefore | \sigma_{s,m} | + | \sigma_{s,b} | \leq 1.5 \cdot S_s$
 $\therefore | 30.6 | + | -75.4 | \leq 207 \text{ N./mm}^2$
 $\therefore 106, \text{ must be } \leq 207 \text{ N./mm}^2$

Computations completed for ASME tubesheet configuration: b

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	144.8 <=	207.0 N./mm ²	Pass
Tubesheet Shear Stress	4.2 <=	110.4 N./mm ²	Pass
Maximum Tube Stress	9.8 <=	92.4 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-8.4 <=	-61.3 N./mm ²	Pass
Maximum Force on any one Tube	153.7 <=	1454.6 Kg _f	Pass
Axial Membrane Stress in Shell	30.6 <=	138.0 N./mm ²	Pass
Shell Stress (jt. inter.)	106.0 <=	207.0 N./mm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm.)	Required	Actual	P/F
Tubesheet Thickness :	74.194	90.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.333	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.333	2.110	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses				Case Type	Pass/Fail
			Bend	Allwd	Shear	Allwd		
D1uc	61.729	48.969	110	207	4	110	Ps+Pt-Th	D1 Ok
D2uc	37.575	...	49	207	8	110	Ps+Pt-Th	D2 Ok
D3uc	74.194	...	145	207	4	110	Ps+Pt-Th	D3 Ok
O1uc	21.772	...	96	459	4	110	Ps+Pt+Th	O1 Ok
O2uc	39.036	...	131	459	11	110	Ps+Pt+Th	O2 Ok



03uc	37.590	...	126	459	4	110	Ps+Pt+Th	03	Ok
04uc	27.578	...	88	459	3	110	Ps+Pt+Th	04	Ok
D1c	66.112	48.969	120	207	4	110	Ps+Pt-Th-c	D1	Ok
D2c	36.743	...	45	207	8	110	Ps+Pt-Th-c	D2	Ok
D3c	76.810	...	153	207	5	110	Ps+Pt-Th-c	D3	Ok
O1c	35.320	...	111	459	4	110	Ps+Pt+Th-c	O1	Ok
O2c	45.474	...	143	459	12	110	Ps+Pt+Th-c	O2	Ok
O3c	45.202	...	142	459	5	110	Ps+Pt+Th-c	O3	Ok
O4c	36.596	...	102	459	4	110	Ps+Pt+Th-c	O4	Ok

Max: 76.8099 48.969 mm. 0.737 0.107 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses ----				:	---- Shell Band Stress ----				: Pass
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd	
D1uc	16	137	...	...	:	...	...	...	...	Ok
D2uc	15	137	...	...	:	...	...	...	...	Ok
D3uc	31	137	...	...	:	...	...	...	...	Ok
O1uc	16	484	...	...	:	...	...	...	...	Ok
O2uc	20	484	...	...	:	...	...	...	...	Ok
O3uc	31	484	...	...	:	...	...	...	...	Ok
O4uc	6	484	...	...	:	...	...	...	...	Ok
D1c	18	137	...	...	:	...	...	...	...	Ok
D2c	16	137	...	...	:	...	...	...	...	Ok
D3c	34	137	...	...	:	...	...	...	...	Ok
O1c	18	484	...	...	:	...	...	...	...	Ok
O2c	23	484	...	...	:	...	...	...	...	Ok
O3c	34	484	...	...	:	...	...	...	...	Ok
O4c	7	484	...	...	:	...	...	...	...	Ok
Max RATIO	0.249				:	...				...

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses ----				Tube Loads		Shell Stress		Channel Stress		Pass
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	
D1uc	24	92	-24	-92	373	1454	237	484	...	...	Ok
D2uc	17	92	-7	-61	261	1454	160	206	...	...	Ok
D3uc	10	92	-8	-61	154	1454	106	206	...	...	Ok
O1uc	24	184	-24	-92	373	2909	237	484	...	...	Ok
O2uc	11	184	-4	-61	168	2909	52	484	...	...	Ok
O3uc	10	184	-8	-61	154	2909	106	484	...	...	Ok
O4uc	7	184	-7	-61	110	2909	119	484	...	...	Ok
D1c	24	92	-24	-92	377	1454	258	484	...	...	Ok



D2c	17	92	-7	-61	272	1454	178	206	...	...	Ok
D3c	11	92	-9	-61	172	1454	113	206	...	...	Ok
O1c	24	184	-24	-92	377	2909	258	484	...	...	Ok
O2c	12	184	-4	-61	182	2909	59	484	...	...	Ok
O3c	11	184	-9	-61	172	2909	113	484	...	...	Ok
O4c	7	184	-7	-61	117	2909	132	484	...	...	Ok

Max RATIO 0.259 0.259 0.259 0.858 ...

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm.)	Required	Actual	P/F
Tubesheet thickness :	76.810	90.000	Ok
Tubesheet thickness flanged extension :	48.969	54.000	Ok
Tube thickness :	0.601	2.110	Ok
Tube-tubesheet fillet weld leg :	1.333	2.110	Ok
Tube-tubesheet groove weld leg :	1.333	2.110	Ok

Min Shell length of thk, (20.000) adj. to tubesheet: 281.169 mm.

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm.
 Due to Pressure Shell Compresses by : -0.087 mm.
 Due to Pressure + Thermal Shell Compresses by : -0.087 mm.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
Tubesheet Bending Stress	41.741	0.997	84.400	1.000
Tubesheet Shear Stress	240.385	1.000	258.135	1.000
Tube Tensile Stress	68.701	1.000	106.952	1.000
Tube Compressive Stress	61.280	1.000	164.415	1.000
Tube-Tubesheet Joint load	68.701	1.000	106.952	1.000
Shell Stress (Axial, Junction)	39.423	0.999	54.479	1.000
Tube Pressure Stress	164.430	1.000	106.120	1.000
Tubesheet Extension Stress	35.494	...	No Calc	No Calc
Minimum MAWP	35.494		54.479	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	0 shellside Stress Rat.	Shellside MAPnc	0 tubeside Stress Rat.
Tubesheet Bending Stress	42.355	1.000	84.400	1.000
Tubesheet Shear Stress	299.311	1.000	272.754	1.000
Tube Tensile Stress	70.025	1.000	111.565	1.000
Tube Compressive Stress	70.025	1.000	193.760	1.000
Tube-Tubesheet Joint load	70.025	1.000	111.565	1.000
Shell Stress (Axial, Junction)	42.355	1.000	64.805	1.000
Tube Pressure Stress	164.430	1.000	106.120	1.000
Tubesheet Extension Stress	35.494	...	No Calc	No Calc
Minimum MAPnc	35.494		64.805	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm.)

Load Case#	Due to Pressure UnCorroded	Due to Pressure Corroded	Due to Thermal Exp. UnCorroded	Due to Thermal Exp. Corroded	Computed UnCorroded	Delta Ms Corroded
1	0.0285	0.0286	...	...	...	...
2	-0.0874	-0.0874	...	...	...	...



3	-0.0588	-0.0589	...	...	...	...
4	...	...	...	...	...	...
5	0.0285	0.0286	...	...	...	...
6	-0.0874	-0.0874	...	...	...	...
7	-0.0588	-0.0589	...	...	...	...
8	...	...	...	...	...	...

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):
 = max(tubesheet thk/4, min(tubesheet thk, shell thickness))
 = max(90/4, min(90, 20))
 = 22.500 mm.

Thickness Ratio = 0.43, Temperature Reduction per Fig. UCS 66.1 = 43 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-4 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-21 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Channel Side MDMT calculation:

Governing thickness for the channel side:
 = tubesheet thickness/4
 = 90/4
 = 22.500 mm.

Thickness Ratio = 0.43, Temperature Reduction per Fig. UCS 66.1 = 43 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-4 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-47 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

where the MDMT reduction ratio per UCS 66 (b)(1)(b) is:
 = max(pt/Tubeside MAPnc, ps/Shellside MAPnc), must be ≤ 1
 = max(15.4/35.5, 20/64.8)
 = 0.434

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